PCB Axial-Flux Motor: Modelling Timeline and Methodology Record

Oct 8, 2026 · @Olwethu

How to use this record

This is the working record of the Maxwell and MATLAB modelling of the PCB-stator axial-flux motor, from the handover on 4 Oct 2026 (about 05:02) to 8 Oct 2026 at 05:52, when the electromagnetic modelling was declared complete. The thermal model is a separate workstream (on RedLab19 from about 04:30 on 8 Oct) and is not covered here. It is written so that the Methodology, Results, Discussion and Recommendations chapters can be drafted from it directly. The section "Using this record in the report" maps each part to the report.

Roles. Olwethu set the engineering specification and made every engineering decision, advised by an external reviewer. Claude did the execution: it wrote and maintained the IronPython automation for ANSYS Electronics Desktop (AEDT) 2025 R2, the post-processing, and the MATLAB model. Claude raised any conflict with the specification instead of changing it.

Provenance tags on every number:

Tag	Meaning
[AEDT]	Solved in Maxwell
[Py]	Computed in Python from AEDT data, or a geometry check
[Est]	Engineering estimate
[Assump]	Modelling assumption

Sources. Lab-PC run logs and status files, which carry PC timestamps and are the most reliable. Then the clock times Olwethu reported, the reviewer's notes, and the project docs. All times are SAST (UTC+2). A time that was not recorded is marked as such and never guessed.

Do not report the Rev1 numbers K = 38.05 (c = 0.866) or K = 56.91 (f = 1.0724) if you find them in the chat. They came from synthetic test data used to check the code.

Superseded headline values. The board-by-board values quoted as headlines until 05:49 on 8 Oct are now cross-checks only. They are Rev0 18.05, Rev1-S 38.5 and Rev1-D 56.8 mN·m/A, and the MATLAB rows built on them (Rev0 27.1 mN·m per A_pk, 1.336 V rms, Km 0.0188; Rev0 3127 rpm and 64.7 %). They are about 6–8 % low, because their single-board reference runs had the region boundary only 5 mm below the board. The section "Full-motor field-only FEM and the validation gate" explains this. Report them only as the reduced-method comparison.

Final electromagnetic results

The electromagnetic modelling closed at 05:49 on 8 Oct, when the validation gate passed. The full-motor field-only model reproduces a detailed single-board FEM with meshed copper to +1.40 % in K1 and 0.04° in phase (limits 2 % and 1°). All values are per phase at g = 1.0 mm on every active gap, with 2.0 mm steel, from field-only FEM with trace-resolved winding post-processing, 20 mm region padding [AEDT field + Py].

Design	Magnets	K (mN·m/A)	T per A_pk, ideal 3-phase (mN·m)	E1 at 1000 rpm (V rms, phase)	R per phase (Ω)	Km (N·m/√W)	H3/H1	K vs Rev0	Board-by-board K (cross-check)	Full-motor vs board-by-board
Rev0	16	19.50	29.2	1.444	1.389	0.0203	5.9 %	1.00 ×	18.05	+8.0 %
Rev1-S	32	41.64	62.5	3.083	2.777	0.0306	5.6 %	2.14 ×	38.54	+8.0 %
Rev1-D	48	60.28	90.4	4.464	2.777	0.0443	5.8 %	3.09 ×	56.89	+6.0 %

- **Stator split.** Rev1-S is S1 19.50 + S2 22.135 mN·m/A. Rev1-D is S1 19.52 + S2 40.76 mN·m/A. In both designs S2 is in phase with S1 (0.0°), so S2 is wired in series in the same sense as S1.
- **Phase balance.** Phases A, B and C agree within 0.25 % in every design. Their phases are −30.1°, +90.0° and −149.9° electrical.
- **Uncertainty.** The field-only model reads 1.4 % above the detailed reference. The likely cause is that B is sampled on the trace centre-lines rather than averaged over the copper cross-section. If a conservative value is wanted, divide by 1.014: 19.23 / 41.06 / 59.45 mN·m/A. No conclusion changes. For Rev1 there is one more open point: the field-map S2 boards read 3–11 % above the detailed Rev1 single-board runs, and the cause is not yet known. The Rev1 absolute values are therefore less certain than Rev0's, while the design ratios agree within 1–2 % between the two methods (see "Full-motor field-only FEM and the validation gate").

- **Other FEM results used in the design.** R per board-phase is 0.2777 Ω [AEDT]. Rev1-D's shared R1 plate peaks at 1.23–1.30 T, so it is unsaturated. R1 and R2 attract each other with about 47–48 N. This force comes from Maxwell stress and is an estimate.
- **System figures** (speed, efficiency, acceleration) come from the MATLAB model and are in the section "MATLAB system model".

Project setup

The Rev0 motor is a single-sided axial-flux machine: a coreless five-PCB stator, a 1.0 mm air gap, and a 16-magnet rotor on a steel back plate. Everything below was modelled against this specification, which Olwethu set on 4 Oct 2026.

Item	Rev0 specification
Stator	5 identical Gould mini_wave-bb PCBs, stacked in the same orientation, mask to mask (pitch 1.02032 mm = 1.0 mm board + 2 × 0.01016 mm solder mask)
PCB	12-sided, about 93 mm across flats, 25 mm opening, 6 × 1 oz (35 µm) copper layers, 615 vias (205 per phase), 148 copper islands per phase
Terminals	In = layer L1, Out = layer L6 at the same XY. A (−22.75, 39.404), B (0, 45.5), C (22.75, 39.404) mm
Winding	Per phase, boards 1→5 in series, additive; three-phase star (star point left open in the models)
Rotor	16 × N35 magnets, 20 (radial) × 6 × 2 (axial) mm, alternating ±Z, centre radius 28 mm; steel_1010 back plate 86/19 mm; 19 mm shaft
Phase sequence	A 0°, B −120°, C +120° electrical (B = A rotated −30° mech, C = A rotated −60° mech)

Tools. ANSYS Electronics Desktop / Maxwell 3D 2025 R2, driven by an IronPython 2.7 pipeline (`main.py` plus modules) that Claude wrote. Post-processing in plain Python. System model in MATLAB, cross-checked in Python and GNU Octave.

Hardware. University lab PCs with about 16 GB RAM (15.8 GB usable) and an 18 GB pagefile:

- GreenLab16 for the single-board gates (4–5 Oct);
- RedLab57 for the five-board stator build (6 Oct);

- RedLab17, RedLab18 and RedLab19 (PC1, PC2, PC3) for the DOE, the final builds and Rev1 (7–8 Oct).

Folder scheme (v8.3). All scripts and finished results sit in one OneDrive folder, `EEE4022S Ansys files\Motor run`, with `rev0_v9\`, `results\` and `frozen\` inside it. This makes every lab PC see the same files. Projects are solved on the PC's local disk (`C:\Users\Student\Rev0_work`), never inside OneDrive, because sync locks the multi-GB solver files. After each run, the small results folder (summaries, CSVs, logs) is copied to `results\` automatically.

Pipeline stages. project → stator → rotor → region_band → excitation → solver_setup → validate → analyze → export. A checkpoint is saved after every stage, so a crashed run resumes from the last good stage.

Pipeline versions.

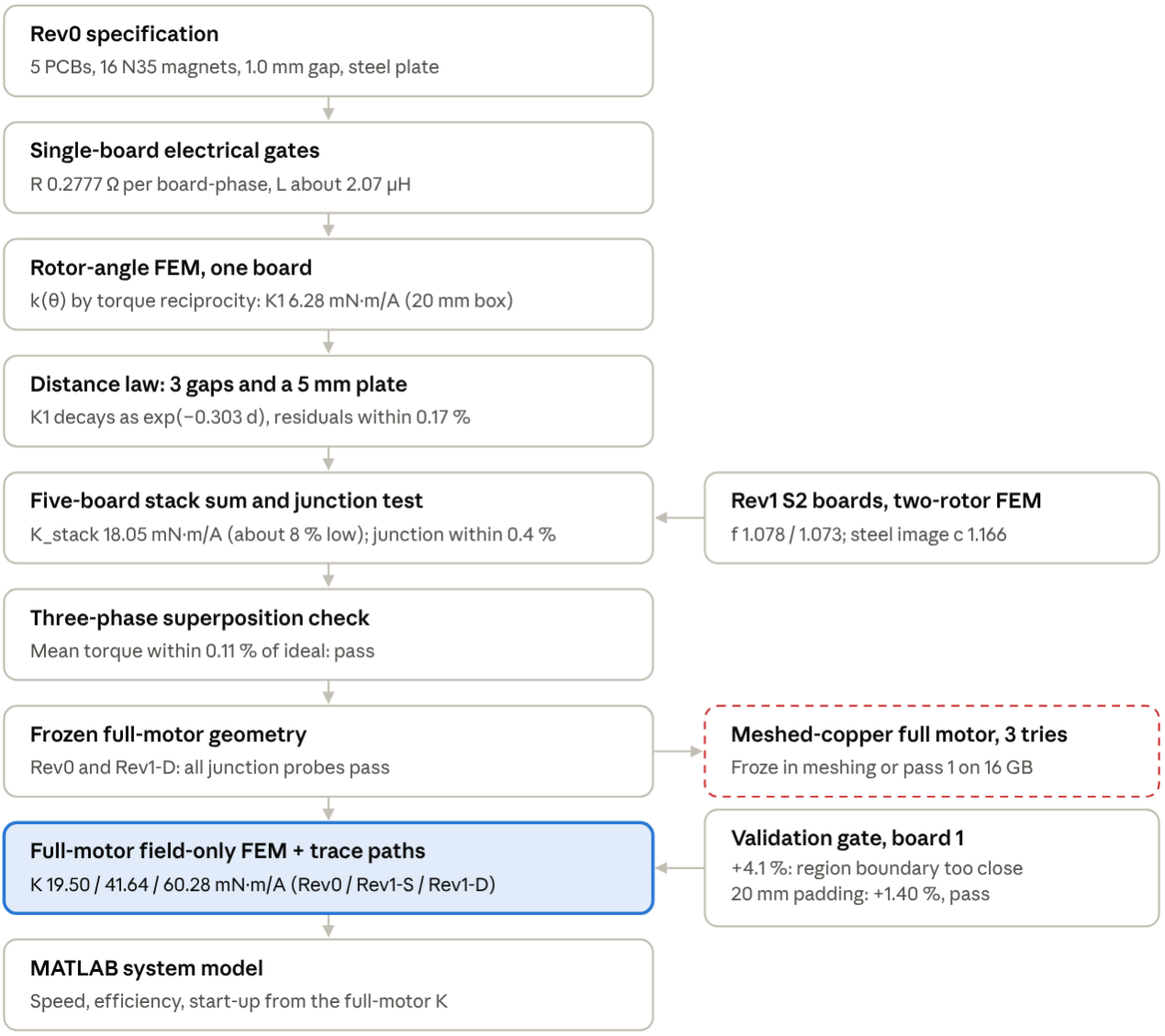
Version	Date	Main change
v1–v3	4 Oct	Gates, dual Gate 2 (T-Omega / A-Phi), checkpoints, solve-acceptance gate, united shaft
v5–v5.2	4–5 Oct	Variables before windings (0 A bug), no project overwrite, inductance matrix off by default
v6–v6.6	5–6 Oct	Rotor-angle magnetostatic method, `rot_angle` variable, stepwise restartable sweep
v7–v7.5	6–7 Oct	Board-by-board five-board stator build, junction probes, region +Y padding guard
v8–v8.3	7 Oct	Batch runner, `stack_sum.py`, reduced DOE, OneDrive / local-disk scheme
v9–v9.3	7 Oct	Final builds from the frozen stator, full-motor attempt, IronPython parser and runner fixes
v9.4	7 Oct, 23:05	Rev1-D (48 magnets): lower PM rotor, per-gap mesh, plate and force field metrics
v9.5	8 Oct, 01:36	Full-motor field-only model (fieldmap.py): magnets, steel and air only; B exported on the ODB++ trace centre-lines (traces_board.json); Lorentz torque in Python; restartable through fieldmap_progress.json; built-in 2 % / 1° validation gate

Version	Date	Main change
v9.5 runner packs	8 Oct, 03:05 and 04:24	Mesh-convergence runners (zone mesh 0.55 mm; D500 gap mesh 1.0 mm), then region-boundary runners (REGION_PADDING 20 mm for the three field maps and the single-board reference)

Chronological timeline

The work ran in fourteen phases from the handover on 4 Oct to 08:37 on 8 Oct, after a Phase 0 on 3 Oct. Each phase removed one risk before the next phase relied on it: conduction first, then the field and torque method, then the stack, then the motor.

Each modelling level was validated before the next relied on it



validation chain · 9 levels, 3 side branches

Read top to bottom: each box was checked before the next box used it. The dashed branch is the meshed-copper full-motor solve that the 16 GB PCs could not complete. The field-only full-motor model (highlighted) replaced it on 8 Oct. It passed its gate only after the region boundary was moved to 20 mm, and its K values feed MATLAB.

Phase 0 — Before the handover (3 Oct)

- Claude built the six-layer stator natively in AEDT from the ODB++ source package (`BUILD_PCB_STATOR_MAXWELL.py`).
- **First full transient run.** The build took about 19 min. The initial mesh was about 1.93 M tetrahedra, took 32 min and peaked at 10.4 GB.
- **It failed** at the first time step: *"Error in calculating the conduction paths. The most likely reason is that the mesh of the source conductor is wrong."*

- **Fix adopted:** boundary terminals. Raised 1 × 0.5 mm leads, 0.3 mm off the board, run out to the region's +Y face with +Y padding 0.
- **Geometry facts fixed here:** 25/25/25/24/24/25 islands per layer (148 per phase); 615 unique vias; minimum spacing between phases 0.241 mm; L2/L3 and L4/L5 carry identical copper in parallel.

Phase 1 — Handover and specification (4 Oct, about 05:02)

- Olwethu reset the roles: *"You are handling the execution and automation in AEDT/Maxwell … Your role is to implement it accurately and robustly in AEDT, identify implementation problems when they occur, and automate the solution without silently changing the engineering specification."*
- The Rev0 specification (see Project setup) became the source of truth, together with the PCB source package.
- **Two conflicts were raised and resolved:**
  - The raw data had 617 via records against 615 vias. Two were near-coincident Phase A duplicates, about 5 µm apart.
  - The shaft changed from 8 mm to 19 mm.

Phase 2 — Pipeline design and review (4 Oct, 18:56–19:27)

- The reviewer audited REV0_PIPELINE and reproduced the geometry checks: 615 vias, 12 NPTH holes, the 25 mm bore, zero cross-phase collisions, the 1.02032 mm pitch and a 1.00000 mm gap.
- **Architecture corrections:**
  - In AEDT 2025 R2, the A-Phi formulation does not support motion. So rotating models use T-Omega, and A-Phi is for stationary checks only.
  - Resume must be failure-safe, with a checkpoint after each stage.
  - A solve is only accepted if it finishes without error-level messages.
  - The shaft is modelled on the rotor side only and united with the back plate.
- **Gate order agreed:** Gate 1 magnetostatic (MS) → Gate 2A T-Omega → Gate 2B A-Phi → Gate 1 for Phases B and C → reference board.

Phase 3 — Single-board electrical gates (GreenLab16, 4 Oct 19:37 → 5 Oct 04:47)

- **Gate 1, Phase A** (MS, 1 A DC, 2 passes): R_A = 0.27776 Ω in about 5 min. PASS [AEDT].
- **Gate 2A, T-Omega, long form:** ran out of 16 GB after one time step. The short form (2 steps, 5 mm padding, no fields) solved in 55 min, but every output was 0 A. Root cause: the design variables were created after the windings. Fixed in v5.

- **Gates 1B and 1C:** R_B = 0.27736 Ω (7.9 min) and R_C = 0.27789 Ω (6.7 min). PASS. The three phases agree within 0.19 %; mean 0.2777 Ω, which is +2.9 % on the 0.2699 Ω reference.
- **Gate 2A, v5:** current 70 mA against 69.76 mA requested; flux linkage gives L ≈ 2.08 µH and induced voltage gives 2.07 µH (they agree within 1 %). PASS [AEDT]. The PC then froze at "Start Matrix" (solver about 6.2 GB), but the results survived a hard power-off.
- **BOARD_3PH_MS** (three phases, FR-4, matrix on) froze at pass 3 "Start Matrix" (8.04 GB, 72 % memory) after about 2 h 20 min. `postprocess_ms.py` recovered the last pass: R 0.27644 / 0.27608 / 0.27648 Ω; L − M ≈ 1.91 µH.
- **BOARD_3PH_MS_AIR:** the same model with FR-4 and solder mask set to air, run with Olwethu's approval on 5 Oct. Resistances changed by ⩽ 0.11 % and stored energy by −0.22 %. PASS.
- **Reviewer, 05:06:** the single board is electrically validated; the terminal architecture is frozen; Gate 2B (A-Phi) is deferred and was never run.

Phase 4 — Rotor-angle reference model (5 Oct 16:26 → 6 Oct 05:05)

- **Method.** One board plus the real rotor, solved magnetostatically at a series of rotor angles. Torque reciprocity gives dλ/dθ = (T(+I) − T(−I))/2I, which is both the torque constant and the back-EMF shape.
- **Smoke run** (2 angles × ±1 A):
  - Failed once because "theta" is a reserved AEDT name; the variable was renamed `rot_angle`.
  - Failed again on the Optimetrics Analyze call; fixed.
  - Then solved in 81 min (about 20 min per angle): K1 = 5.81 mN·m/A, ψ −23.8°. Treated as preliminary.
- **Full-period baseline** (0–45° in 5.625° steps × ±1 A = 18 solves):
  - The lab PC crashed at 22:21, and Optimetrics saves only at the end.
  - v6.6 introduced the stepwise sweep: the project and `sweep_progress.json` are saved after every solve.
  - Resumed at 02:09 on 6 Oct; finished 05:05.
- **Baseline result [AEDT]:**
  - K1 = 6.0917 mN·m/A, ψ = −30.18° electrical, λ1 = 0.7615 mWb, E1 = 0.451 V rms per board at 1000 rpm.
  - 3rd harmonic H3/H1 = 11.75 %.
  - Quality: full-period repeat error 0.019 %; half-wave antisymmetry error 0.051 %; rotor-vs-copper torque mismatch ⩽ 0.116 %; copper-torque nonlinearity 0.005 %.
  - Steel |B|max 1.39–1.44 T; mid-gap mean |Bz| 0.168 T.

Phase 5 — Five-board stator build (6 Oct 16:48 → 7 Oct about 02:30)

- **Reviewer:** bridge each board's L6 Out to the next board's L1 In, and do not make the star point yet.
- **v7 first attempt.** One united solid per phase across all five boards. It ran for more than 2.5 h, then AEDT closed. This led to v7.1–v7.3: no cross-board union, history purged per board, per-board checkpoints, and a stator-only build (Olwethu's workflow).
- **REV0_STATOR_BUILD on RedLab57:**
  - Run 1 failed on expected build-only notices; fixed in v7.4.
  - Run 2 built the boards one at a time. Cumulative times: 406 / 1148 / 2543 / 5280 / 9733 s for boards 1–5, slowing as the model grew.
  - 12 copper bridges (4 per phase); all 36 junction body probes passed.
- **Validate failed at 00:15:49 on 7 Oct:** *"Illegal external terminal 'PhaseC_In': An external terminal must border the edge of the problem region…"*, for all six terminals. Root cause: the region's +Y padding was 5 mm.
- **Olwethu set +Y padding to 0 by hand**, validated, and saved `REV0_STATOR_FROZEN.aedt`. The v7.5 guard now forces this automatically.
- **v8/v8.1 (about 00:30–02:30).** Reduced DOE plan, `stack_sum.py`, and an analytical preview: a 5-board stack gives about 3× one board, not 5×.

Phase 6 — DOE, Rev0 decision, final build (7 Oct 13:03 → 16:47)

- **13:03.** First Rev1 specification locked (S1–R1–S2–R2), plus the OneDrive / local-disk scheme (v8.3).
- **Runs:**
  - REV0_DOE_D050 (0.5 mm gap): K1 7.11 mN·m/A, 72 min.
  - REV0_DOE_D500 (5.0 mm gap): K1 1.815 mN·m/A, about 11 min per angle.
  - STACK_JUNCTION_TEST (2 boards): 0.55313 Ω, PASS.
  - REV0_DOE_BI50 (5 mm plate): K1 6.097 mN·m/A.
- **Distance law fitted [Py on AEDT]:** $K1(d) = 6.1006 \cdot e^{(-0.3031(d-1))}$ mN·m/A, residuals ⩽ 0.17 %. At g = 1 mm the five-board stack gives K_stack = 18.05 mN·m/A.
- **16:15 — Olwethu locked Rev0 at g = 1.0 mm and t = 2.0 mm.**
- **REV0_FINAL_G100_T20** built from a copy of the frozen stator in 13.7 min: 36/36 probes, gap 1.0000 mm. Frozen.

Phase 7 — Three-phase gate and full-motor attempts (7 Oct 16:54 → 23:02)

- **17:23 — two-PC plan accepted.** PC1 runs the reduced (authoritative) method. PC2 deliberately attempts the full five-board solve, so the failure can be documented.

- **The three-phase single-board model crashed RedLab17 at about 18:14.** The solver had reached 8.64 GB, the PC was at 95 % memory, and it was still on pass 1, "Start Matrix".
  - Two pipeline bugs were fixed along the way: a baseline summary without `fit_coeffs`, and IronPython's literal parser rejecting negative numbers.
- **Three full five-board attempts:**
  - Attempt 1 froze at 18:08.
  - Attempt 2 froze at about 19:00 in initial meshing ("Fixing local region mesh issues"; the mesher alone used 5.5 GB).
  - Attempt 3 started at 21:13, began solving at 21:24:59 with only 4.03 GB free, and froze at 23:02.
- **The three-phase gate was replaced by superposition.** Phases B and C were each solved alone (REV0_CHECK_PHB on RedLab17, REV0_CHECK_PHC on RedLab18), and `phase_check.py` summed them.
- **Result: PASS at about 22:31.** Mean synchronous torque 9.1478 mN·m per A_pk against the ideal 9.1375 (+0.11 %).

Phase 8 — MATLAB system model (7 Oct 21:20 → 22:41)

- Olwethu proposed modelling the motor in MATLAB. The reviewer endorsed it as a complementary FEM-informed lumped model, not a fallback.
- Models 1–3 were run at 22:15–22:20, then re-run at 22:34 with the measured B/C phase corrections. Every change was ⩽ 0.3 % (results in "MATLAB system model").
- **22:41.** Rev0 declared closed at three levels: detailed FEM, reduced five-board model, and MATLAB model.

Phase 9 — Rev1 becomes two designs (7 Oct 22:48 → 8 Oct)

- **22:56.** The reviewer raised a conflict: in the 32-magnet Rev1, S2 faces bare steel on one side.
  - New primary design **Rev1-D**: 48 magnets, with a double-sided middle rotor.
  - Comparator **Rev1-S**: the old 32-magnet design.
- **23:05.** Pipeline v9.4 delivered.
- **Runs:**
  - REV1D_FINAL built and frozen on RedLab19 (23:28 → about 00:17).
  - REV1D_S2_K3 started 23:23 on RedLab18; 26.8 min per angle.
  - REV1D_S2_K1 froze at 23:44 and was restarted at 00:28.
  - REV1_S2_ENV (Rev1-S) started at 00:26 on RedLab19.
- **00:38.** MATLAB extended to Rev1-S and Rev1-D (see "MATLAB system model").

Phase 10 — Rev1 board-by-board FEM finished; full-motor method chosen (8 Oct 00:26 → 01:48)

- **00:26.** RedLab19 (PC3) started REV1_S2_ENV, the Rev1-S S2 board next to R1′s bare steel. **00:28:** RedLab17 (PC1) restarted REV1D_S2_K1 after a clean reboot; its first attempt froze at 23:44.
- **00:38. Olwethu asked four questions:**
  - Is the construction of the motors complete?
  - What in Maxwell consumes the RAM?
  - Would the full motors solve with 32, 64 or 128 GB, and how long would each take?
  - Would a hand-built model run where the scripted one did not?
  - He also asked for Rev1 in MATLAB and for this record.
- **00:40. Claude′s answer** [Est]:
  - Construction is complete. The Rev0 and Rev1-D full motors are built, frozen and probe-checked. The Rev1-S build takes about 40 min whenever a PC is free.
  - The RAM goes to three things: the mesh around 35 µm copper with 20 µm board-to-board gaps, the direct-solver matrix, and the adaptive passes.
  - 32 GB will not solve either motor. 64 GB is borderline for Rev0. 128 GB should solve Rev0 at roughly 4–24 h per angle. Rev1 needs 128–256 GB or HPC.
  - A hand-built model is the same model, so it would not help.
  - Four ways round the limit: the iterative solver, equivalent conductors, a sector model, or a large workstation or HPC.
- **Run progress:**

Time	PC	Run	Progress
00:51	PC2	REV1D_S2_K3	Variation 3/4, 27.1 min
00:57	PC1	REV1D_S2_K1	Variation 1/4, 26.3 min
01:04	PC3	REV1_S2_ENV	Variation 3/4, 11.4 min
01:20	PC1	REV1D_S2_K1	Variation 2/4, 22.5 min
01:42	PC1	REV1D_S2_K1	Variation 3/4, 21.1 min

- **00:59.** MOTOR_MATLAB_MODEL.zip delivered: Rev0, Rev1-S and Rev1-D through Models 1–3, plus compare_designs. **01:02:** the first version of this record was published.

- **01:10. Olwethu chose alternative 2,** equivalent conductor blocks, for Rev0 and Rev1. **01:12:** Claude reported that the ODB++ files hold every trace centre-line, so the exact current path could replace the blocks, and asked which version to build. Olwethu answered "Essentially option 1": traces as line currents.
- **The reviewer endorsed option 1** (before 01:36; exact time not recorded). The reviewer's conditions:
  - Use the actual 3-D centre-lines of every PCB trace and via.
  - Maxwell solves only the magnets, the steel and the air, at prescribed rotor angles.
  - Integrate dF = I·dl × B in Python, and obtain $e_x = \omega \cdot k_x(\theta)$ by reciprocity.
  - Validate on one Rev0 board first: it must reproduce K1 ≈ 6.09 mN·m/A within 2 % before the five-board or Rev1 results are used.
  - Do not call it "exact". The name is "field-only FEM with trace-resolved winding post-processing".
  - Solid meander blocks were rejected because they would need an equivalent width, turns, layer split, via behaviour and axial placement, all calibrated against FEM anyway.
- **01:33.** Olwethu ran MATLAB run_all with the pre-FEM Rev1 inputs. Results: Rev0 18.02, Rev1-S 36.04 (lower bound) and Rev1-D 54.06 mN·m/A (pure superposition).
- **About 01:40.** K3 (RedLab18) and S2_ENV (RedLab19) finished [AEDT]:
  - **K3, the middle S2 board of Rev1-D:**
    - K1 = 7.0504 mN·m/A, ψ −30.04°, H3/H1 4.49 %.
    - The prediction was 6.5731, so f = 1.0726.
    - The two rotors' fields add, so the "through" magnetisation works.
    - S2's phase matches S1's within 0.1°.
  - **S2_ENV:** K1 = 2.5970 mN·m/A, ψ −30.05°. That is ×1.467 on the free value at 5.081 mm (1.7707), so c = 1.166.
  - **K3 field metrics:**
    - R1 plate |B|max 1.23–1.25 T; R2 steel 1.42–1.47 T.
    - Axial force on R2 about −47.8 N.
    - The 11.25° row is blank because the field export failed at that angle.
- **01:41.** Claude gave the MATLAB inputs: c_image 1.166, and fD_K1 = fD_K3 = 1.0726 (K1's value provisional). **01:48:** MATLAB rerun gave Rev1-S 38.48 and Rev1-D 56.68 mN·m/A.

Phase 11 — Field-only full-motor model: build, first runs, gate failure (8 Oct 01:36 → 03:17)

- **01:36. Pipeline v9.5 delivered** (Motor_run_rev0_v9_5.zip, which replaces the whole rev0_v9 folder). It contains:

- **The trace data** (traces_board.json, built offline from the ODB++ export):
  - Lines and arcs on all six copper layers are split into 0.4 mm chords. Each chord is assigned to its net-correct copper island.
  - Pads join the nodes inside them. Vias form a chain through the six layers. The terminals sit on the L1 and L6 pads.
  - A DC nodal solve at 1 A gives the current in every piece.
  - Per board-phase that is about 11,040 track pieces and 1,015 via steps, with R_dc = 0.2656 Ω. The three phases agree within 0.06 %; the meshed-copper FEM gave 0.2777 Ω.
  - The layer heights on board 1 are ±0.48222, ±0.301015 and ±0.12009 mm. Each further board is offset by the 1.02032 mm pitch.
- **The Maxwell model:**
  - Magnets, steel and air only, with no copper.
  - Air zones (r 13.5–47.5 mm) around each stator, meshed at 0.8 mm or finer, up to 500,000 elements. Magnets meshed at 2.0 mm.
  - 4 adaptive passes, 20 % refinement, inductance matrix off.
  - 8 rotor angles, 0–39.375° mechanical in 5.625° steps, which is one electrical period.
- **The field export:**
  - B is exported on 180,775 points for Rev0 or 361,550 for Rev1, using FieldsReporter.ExportToFile with a points file.
  - Four call variants are tried on 20 probe points first, and the one whose echoed points match is kept.
- **The post-processing:**
  - Lorentz torque per piece, then k = −T/I.
  - Fourier fit for H1 and H3.
  - Results per board, per stator and per machine, the S2-vs-S1 phase, and the gate line.
  - Outputs: fieldmap_summary.txt, fieldmap_results.json and fieldmap_k_theta.csv.
- **Offline check before delivery** [Py]. The same trace calculation was run with a hand-calculated magnet field: surface charges plus an ideal-iron image of the back plate. It gave K1 = 6.110 mN·m/A at −30.2°, against the detailed FEM's 6.0917 at −30.18°. The torque at each angle agreed within about 1 %. This confirmed the trace extraction, the current directions and the signs.

- **02:02. PC1 crashed during K1's fourth angle.** After the restart, the lab PC had wiped its local work folder (C:\Users\Student\Rev0_work), so K1 started again from angle 1. This was found at 02:49.
- **About 02:10–02:49. The first field maps** (5 mm padding, 0.8 mm zones) [AEDT field + Py]:
  - FIELDMAP_REV0 on PC2: 0.8 min per angle, 6.6 min in total.
  - FIELDMAP_REV1D on PC3.
  - The export rejected variant 0 (echoed points did not match) and accepted variant 1 (a "Unit=mm" header in the points file). The probe |B|max was 0.200 T.
- **02:52–03:15.** FIELDMAP_REV1S ran on PC2: 1.5 min per angle, 12.3 min in total.

Field map (5 mm padding)	Board 1 K1	Board 5 K1	S1 stack	S2 stack	Machine K	Gate vs 6.0917
FIELDMAP_REV0	6.3429	1.9405	19.265	—	19.265	FAIL, +4.12 %, −0.02°
FIELDMAP_REV1D	6.3496	1.9468	19.295	40.750	60.045	FAIL, +4.23 %
FIELDMAP_REV1S	6.3337	1.9410	19.259	22.132	41.391	FAIL, +3.97 %

- The Rev1-D S2 boards are symmetric: 8.749 / 7.853 / 7.546 / 7.855 / 8.748 mN·m/A. The Rev1-S S2 boards decay towards R1's steel: 6.649 / 5.184 / 4.090 / 3.327 / 2.882. S2 is in phase with S1 (0.0°) in both designs.
- **03:05. Gate failure reported.** By the reviewer's rule, no field-map number was used.
  - **First hypothesis:** the field-map mesh was too coarse.
  - **Second suspect:** the detailed 5 mm point, D500. Its gap mesh was 2.5 mm (the g/2 rule), and it sat 8 % below the hand-calculated field (1.815 against 1.97).
  - Four finer-mesh runners were delivered.
- **03:17. Claude compared the design ratios.** The two methods agreed within 1 %: Rev1-D/Rev0 3.12 against 3.15, and Rev1-S/Rev0 2.15 against 2.14. The absolute values differed by 4–7 %.

Phase 12 — Diagnosis: mesh ruled out, region boundary found (8 Oct 03:15 → 04:33)

- **03:15. Three runs started:**
  - PC3: FIELDMAP_REV0_M055, with the zone mesh refined from 0.8 to 0.55 mm.
  - PC2: REV0_DOE_D500_M10, the 5 mm single board with its gap mesh refined from 2.5 to 1.0 mm.
  - PC1: K1's variation 2/4 finished (22 min).

- **04:01. The finer field-map mesh changed almost nothing.** Board 1 went from 6.3429 to 6.3408 (−0.03 %), and the stack from 19.265 to 19.262. The gate still failed (+4.09 %). The same message reported that K1 had finished on PC1, at about 04:00.
- **04:02.** Claude concluded that the mesh does not cause the gap and looked for a cause inside the models.
- **Root cause: the outer region boundary** [Py, analytic].
  - Maxwell's default outer boundary forces the flux to run along the region faces (B·n = 0). That acts like a same-sign image of the magnets across the face.
  - Every detailed single-board run used 5 mm padding, so its bottom face sat 5 mm below the board. The field map's region ends 5 mm below the whole five-board stack, about 10 mm below board 1.
  - The hand-calculated field model was run with such a plane added:

B·n = 0 plane	Board 1 K1 (mN·m/A)	Board 5 K1 (mN·m/A)
None	6.1105	1.9227
5 mm below board 1 (single-board models)	5.8570 (−4.1 %)	—
5 mm below the five-board stack (field map, 5 mm padding)	6.0891 (−0.35 %)	1.8563 (−3.5 %)

- **The prediction matched.** The model predicts the field map should read 3.9 % above the single-board reference at board 1; it read 4.1 % above.
- **04:24. Root cause reported, and four padding runners delivered:** FIELDMAP_REV0_PAD20, FIELDMAP_REV1D_PAD20, FIELDMAP_REV1S_PAD20, and REV0_BASELINE_PAD20 (project REV0_DOE_D100_P20). The consequences stated then:
  - Rev0's true K is probably 19.3–19.5 mN·m/A, not 18.05.
  - The far boards are hit hardest, so the fitted distance law (k = 0.3031 /mm) is too steep.
  - Design ratios, the three-phase check and the back-plate choice compare models with the same boundary, so they barely move.
- **04:27. K1 results arrived** [AEDT]. Olwethu asked what the remaining runs were for. He also reported that RedLab19 (PC3) had moved to the thermal model.
  - K1 = 8.4860 mN·m/A, ψ −30.16°, H3/H1 8.93 %. The prediction was 7.8711, so f = 1.0781.
  - R1 plate |B|max 1.27–1.30 T; R2 steel 1.41–1.49 T; mid-gap mean |Bz| 0.148 T.
  - Axial force on R2 −47.2 to −47.3 N.
- **04:28. Claude's answer:**

  - All three designs are fully modelled. The design ratios are robust, because two independent methods agree within 1 %. Only the absolute level was 4–7 % uncertain.
  - Settling the absolute level matters for two reasons: the prototype back-EMF test will be compared against it, and it turns an unexplained 4 % into a documented finding.
  - Stopping there and quoting the board-by-board values with a −0 / +7 % uncertainty would also have been defensible.
  - The minimum plan needed two PCs: PC1 runs the single-board reference with 20 mm padding; PC2 runs the Rev0 field map with 20 mm padding.
  - The final board-by-board stack sums: Rev0 18.05, Rev1-S 38.54, Rev1-D 56.89 mN·m/A. The S2 factors are 1.0781 / 1.0754 / 1.0726 / 1.0754 / 1.0781, giving K_S2 = 38.84.
- **04:33. D500_M10 finished:** K1 = 1.8148 against 1.815 before (−0.01 %), with a gap mesh 2.5 times finer. The mesh is converged in the detailed model as well. That leaves the boundary as the only explanation. Olwethu confirmed the two-PC plan.

Phase 13 — 20 mm padding runs, gate passed, final MATLAB, electromagnetic side closed (8 Oct 04:33 → 05:52)

- **About 04:35. Two runs started:**
  - PC1: REV0_BASELINE_PAD20 (project REV0_DOE_D100_P20). This is the same detailed single board, with meshed copper, at g = 1 mm, solved at 4 angles with +I, but with 20 mm region padding.
  - PC2: FIELDMAP_REV0_PAD20.
- **04:45. FIELDMAP_REV0_P20 behaved as the boundary model predicted** [AEDT field + Py]. Every row below uses the same 0.8 mm zone mesh:

Rev0 field map	5 mm padding	20 mm padding	Change	Predicted by the boundary model
Board 1 K1 (mN·m/A)	6.3429	6.3697	+0.42 %	+0.35 %
Board 5 K1 (mN·m/A)	1.9405	2.0154	+3.9 %	+3.6 %
Stack K (mN·m/A)	19.265	19.496	+1.2 %	—

- **The boundary-free Rev0 values:**
  - Boards 6.3697 / 4.7962 / 3.5881 / 2.7268 / 2.0154 mN·m/A.

  - Phases A / B / C: 19.496 / 19.453 / 19.491 mN·m/A.
  - ψ −30.06°, H3/H1 5.9 %.
  - T 29.2 mN·m per A_pk, E1 1.444 V rms at 1000 rpm, Km 0.0203.
  - The decay through the stack is 0.282 /mm, close to the 2D value π/τ = 0.286 /mm.
  - The summary still printed "FAIL" because its gate compared against the old, suppressed 6.0917.
- **04:45 → 05:05. FIELDMAP_REV1D_P20 on PC2:**
  - S1 rose to 19.523 (from 19.295).
  - S2 stayed at 40.761 (from 40.750, +0.03 %). It lies between two steel-backed rotors, so the region boundary cannot reach it. This was one more confirmation of the boundary explanation.
  - Machine K = 60.284 mN·m/A (B 60.146, C 60.276), with S2 vs S1 at 0.0°.
- **05:05 → 05:17. FIELDMAP_REV1S_P20 on PC2:**
  - S1 19.501; S2 22.135 (from 22.132).
  - Machine K = 41.636 mN·m/A, with S2 vs S1 at 0.0°.
- **05:18. MOTOR_MATLAB_MODEL_v2 delivered.** It sets USE_FIELDMAP = true:
  - K_fieldmap: Rev0 19.496, Rev1-S 41.636 and Rev1-D 60.284 mN·m/A. H3/H1: 5.9 / 5.6 / 5.8 %.
  - The stack factors are rescaled so that each machine's K equals its field-map value. P.K1_boardsum keeps the board-by-board value for comparison.
  - The FEM factors c = 1.166, fD_K1 = 1.0781 and fD_K3 = 1.0726 are built in.
  - Instruction given: do not treat the results as final until the gate passes.
- **05:36.** Olwethu ran run_all with v2. The comparison matched the preview exactly (see the section "MATLAB system model").
- **05:49. PC1 finished REV0_DOE_D100_P20** [AEDT]:
  - K1 = 6.2816 mN·m/A, ψ −30.18°, H3/H1 11.42 %.
  - λ1 = 0.7852 mWb; E1 = 0.465 V rms per board at 1000 rpm.
  - Moving the boundary from 5 mm to 20 mm raised the detailed single board by +3.1 %, from 6.0917.
- **05:50. Gate PASSED.** The field map's board 1 (6.3697, −30.14°) is within +1.40 % and 0.04° of the new reference, against limits of 2 % and 1°.
  - The full-motor field-map values became the headline results. The board-by-board values became the cross-check.
  - MATLAB v2 is final.
  - The project notes were updated: claude/fieldmap_full_motor_method.md and claude/matlab_compare_2026-10-08.md.

- **05:52.** Olwethu declared the electromagnetic side complete. The thermal model continues separately.

Phase 14 — Rev1 S2 check attempted and stopped (8 Oct 06:18 → 08:37)

- **06:18. Two runners delivered** (Motor_run_v9_5_K3_check.zip) to test the Rev1 S2 discrepancy on the middle board (K3):
  - run_REV1D_S2_K3_P20_M10.py: 20 mm padding and a 1.0 mm gap mesh in both gaps (was 1.52 mm).
  - run_REV1D_S2_K3_M10.py: the 1.0 mm gap mesh only, to separate the mesh effect from the padding effect.
- **06:24.** Both started: P20_M10 on RedLab17 (PC1), M10 on RedLab18 (PC2).
- **About 07:00.** PC2 froze; it was switched off. The likely cause is RAM: the finer mesh roughly triples the elements in the two 3 mm gaps of this two-rotor model.
- **About 08:17.** A lighter fallback runner was delivered: run_REV1D_S2_K3_P20.py, with 20 mm padding and the original 1.52 mm gap mesh.
- **08:19.** PC1 froze as well. It is not recorded how many angles had finished.
- **08:37. Olwethu stopped the electromagnetic runs.** The Rev1 S2 discrepancy remains an open point and a stated limitation. It does not change any design conclusion.

Register of Maxwell models and runs

The project made 42 Maxwell runs, numbered 0–41. 30 gave usable results (passed, recovered, done or frozen); 12 failed, were invalid or were not run.

- Every single-phase model of one or two boards solved on 16 GB.
- So did every field-only full-motor model, up to 48 magnets at 1.5 min per angle.
- Every failure involved one of five things: a 1.0 mm gap mesh in the two-rotor model (runs 40–41), three phases with the rotor, the full meshed-copper motor, the inductance-matrix step, or a pipeline bug that was then fixed.
- Runs 31–39 (8 Oct) are the field-only full-motor models and the checks on them.

Abbreviations: MS = magnetostatic; GL16 = GreenLab16; RL = RedLab; "angles" = rotor-angle variations.

#	Project	Purpose	Model and settings	PC	When	Result	Status
0	Native build, transient	First build	Full board, 3 phases, FR-4, rotor; "loop" terminals	Lab PC	3 Oct	1.93 M tets, 10.4 GB; conduction-path error	Failed

#	Project	Purpose	Model and settings	PC	When	Result	Status
1	REV0_GATE1_MS_PHASEA	Conduction, Phase A	1 board, Phase A; MS 1 A, 2 passes, 15 %	GL16	4 Oct 19:37	R_A 0.27776 Ω, about 5 min	Pass
2	REV0_GATE2_TOMEGA_PHASEA	Gate 2A, long	T-Omega, about 5 steps, 15 mm padding	GL16	4 Oct 20:07	RAM exhausted after step 1	Failed
3	REV0_GATE2_TOMEGA_PHASEA_SHORT	Gate 2A, short	T-Omega, 2 steps, 5 mm padding	GL16	4 Oct 21:23–22:19	Solved, but 0 A (v4 bug)	Invalid
4	REV0_GATE1_MS_PHASEB	Conduction, Phase B	As run 1	GL16	4 Oct about 22:30	R_B 0.27736 Ω, 7.9 min	Pass
5	REV0_GATE1_MS_PHASEC	Conduction, Phase C	As run 1	GL16	4 Oct about 22:50	R_C 0.27789 Ω, 6.7 min	Pass
6	REV0_GATE2_TOMEGA_PHASEA_V5	Gate 2A, fixed	T-Omega, 2 steps	GL16	4 Oct 23:31 → 5 Oct 00:26	L ≈ 2.07 µH, I 70 mA; froze afterwards at Start Matrix (6.2 GB)	Pass
7	REV0_GATE2_APHI	Gate 2B	A-Phi	—	—	Deferred by the reviewer	Not run
8	REV0_BOARD_3PH_MS	3 phases, FR-4, inductance matrix	MS, 3 passes, matrix on	GL16	5 Oct 00:44 → about 03:00	Froze at pass 3 (8.04 GB); recovered R, L − M	Recovered
9	REV0_BOARD_3PH_MS_AIR	FR-4 as air check	MS, 3 passes, matrix off	GL16	5 Oct 03:34–04:37	⩽ 0.22 % change	Pass
10	REV0_ROTOR_MOTION	Rotor motion check	Rotor only, T-Omega motion	—	Time not recorded	Rotation correct	Done
11	REV0_REF_ROTOR_MS_SMOKE	Torque reciprocity method	Board 1 + rotor; MS 4 passes; 2 angles × ±1 A	GL16	5 Oct 16:31–18:14	K1 5.81 mN·m/A (preliminary)	Pass
12	REV0_REF_ROTOR_MS_SWEEP	Full-period baseline	9 angles × ±1 A, fields saved	GL16	5 Oct 18:38 → 6 Oct 05:05 (crash 22:21, resumed)	K1 6.0917, ψ −30.18°	Pass
13	REV0_STACK_BUILD (v7)	5 boards + rotor	United solid per phase	—	6 Oct about 17:10	AEDT closed after more than 2.5 h	Failed
14	REV0_STATOR	Five-board stator	5 boards, 3 phases, 12 bridges; build only	RL57	6 Oct 20:59 → 7 Oct 00:15	36/36 probes; terminal padding fixed by hand	Frozen
15	REV0_REF_ROTOR_MS_3PH	3-phase synchronous gate	1 board, 3 phases + rotor; 5 angles	RL17	7 Oct 17:42 → crash about 18:14	8.64 GB at pass 1, 95 % memory	Abandoned
16	REV0_STACK_JUNCTION_TEST	Board-to-board junction	2 boards, Phase A, 1 bridge; MS 1 A	RL18	7 Oct before 14:52 (857 s)	0.55313 Ω (−0.4 %)	Pass
17	REV0_DOE_BI50	5 mm back plate	g 1 mm; 4 angles, fields saved	RL18	7 Oct 14:52–15:53	K1 6.097 (+0.1 %)	Done

#	Project	Purpose	Model and settings	PC	When	Result	Status
18	REV0_DOE_D050	0.5 mm gap	Gap mesh 0.25 mm; 4 angles	RL17	7 Oct, about 72 min	K1 7.11	Done
19	REV0_DOE_D500	5.0 mm gap	Gap mesh 2.5 mm; 4 angles	RL17	7 Oct 15:56–16:44	K1 1.815	Done
20	REV0_FINAL_G100_T20	Final Rev0 geometry	Frozen stator + rotor; build only	RL18	7 Oct, finished by 16:47 (13.7 min)	36/36 probes; gap 1.0000 mm	Frozen
21	REV0_FULL_SOLVE_ATTEMPT 1	Full motor, 1 angle	45 solids; MS 2 passes, matrix off	RL18	7 Oct, froze 18:08	Nothing recorded	Failed
22	REV0_FULL_SOLVE_ATTEMPT 2	As above	As above	RL18	7 Oct, froze about 19:00	Stuck in initial mesh; mesher 5.5 GB	Failed
23	REV0_FULL_SOLVE_ATTEMPT 3	As above	As above	RL19	7 Oct 21:13 → 23:02	4.03 GB free at solve start; no pass completed	Failed
24	REV0_CHECK_PHB	Phase B for superposition	1 board, Phase B + rotor; 4 angles	RL17	7 Oct (4271 s)	K1 6.1107, ψ +90.02°	Done
25	REV0_CHECK_PHC	Phase C for superposition	As above, Phase C	RL18	7 Oct (4811 s)	K1 6.0935, ψ −149.78°	Done
26	REV1D_S2_K1_G100_T20	Rev1-D, S2 board next to R1	1 board between two PM rotors (1.0 / 5.08 mm); 4 angles	RL17	7 Oct 23:23 (froze 23:44); rerun 00:28; crashed 8 Oct 02:02, restarted from scratch after the local-folder wipe; finished about 04:00	K1 8.4860, ψ −30.16°; f = 1.0781; R1 plate 1.27–1.30 T; force on R2 −47.3 N	Done
27	REV1D_S2_K3_G100_T20	Rev1-D, middle S2 board	Both rotors 3.04 mm away; 4 angles	RL18	7 Oct 23:23 → 8 Oct about 01:40 (26.8–27.1 min per angle)	K1 7.0504, ψ −30.04°; f = 1.0726; R1 plate 1.23–1.25 T; force −47.8 N	Done
28	REV1D_FINAL_G100_T20	Final Rev1-D geometry	48 magnets, 2 stators; build only	RL19	23:28 → about 00:17	S1 and S2 36/36 probes	Frozen
29	REV1_S2ENV_G100_T20	Rev1-S, S2 board next to R1 steel	1 board, R2 at 5.08 mm, steel 1 mm below; 4 angles	RL19	8 Oct 00:26 → about 01:40 (11.4 min per angle)	K1 2.5970, ψ −30.05°; c = 1.166	Done
30	REV1_FINAL_G100_T20	Final Rev1-S geometry	32 magnets, 2 stators; build only	—	—	—	Not run

#	Project	Purpose	Model and settings	PC	When	Result	Status
31	FIELDMAP_REV0	Full Rev0, field-only	16 PM + steel + air, no copper; zones 0.8 mm; 8 angles; padding 5 mm	RL18 (PC2)	8 Oct about 02:10–02:49 (6.6 min)	S1 19.265; board 1 6.3429; gate +4.12 %	Done (superseded by 36)
32	FIELDMAP_REV1D	Full Rev1-D, field-only	48 PM, two steel bodies, 10 boards sampled; otherwise as 31	RL19 (PC3)	8 Oct about 02:10–02:49	60.045 (S1 19.295 + S2 40.750); S2 vs S1 0.0°	Done (superseded by 37)
33	FIELDMAP_REV1S	Full Rev1-S, field-only	32 PM; otherwise as 31	RL18 (PC2)	8 Oct 02:52–03:15 (12.3 min)	41.391 (S1 19.259 + S2 22.132); S2 vs S1 0.0°	Done (superseded by 38)
34	FIELDMAP_REV0_M055	Field-map mesh convergence	As 31, zone mesh 0.55 mm	RL19 (PC3)	8 Oct 03:15 → about 04:00	Board 1 6.3408 (−0.03 %); stack 19.262	Done
35	REV0_DOE_D500_M10	Detailed-model mesh check at 5 mm	As 19, gap mesh 1.0 mm instead of 2.5 mm	RL18 (PC2)	8 Oct 03:15 → about 04:30	K1 1.8148 (−0.01 % on 1.815)	Done
36	FIELDMAP_REV0_P20	Boundary test; final Rev0	As 31, padding 20 mm	RL18 (PC2)	8 Oct about 04:35–04:45	19.496 (B 19.453, C 19.491); board 1 6.3697	Done (headline)
37	FIELDMAP_REV1D_P20	Final Rev1-D	As 32, padding 20 mm	RL18 (PC2)	8 Oct 04:45–05:05	60.284 (S1 19.523 + S2 40.761)	Done (headline)
38	FIELDMAP_REV1S_P20	Final Rev1-S	As 33, padding 20 mm	RL18 (PC2)	8 Oct 05:05–05:17	41.636 (S1 19.501 + S2 22.135)	Done (headline)
39	REV0_DOE_D100_P20	Corrected gate reference	Detailed board 1 + rotor (copper meshed), g 1 mm, 4 angles, +I, padding 20 mm	RL17 (PC1)	8 Oct about 04:35–05:49	K1 6.2816, ψ −30.18°; gate PASS (+1.40 %, 0.04°)	Pass
40	REV1D_S2_K3_P20_M10	Rev1 S2 check: padding and mesh	As 27, padding 20 mm, both gap meshes 1.0 mm	RL17 (PC1)	8 Oct 06:24 → froze 08:19	No result recorded	Failed
41	REV1D_S2_K3_M10	Rev1 S2 check: mesh only	As 27, both gap meshes 1.0 mm, padding 5 mm	RL18 (PC2)	8 Oct 06:24 → froze about 07:00	No result; PC switched off	Failed

Software failures and fixes

Thirty-seven failures were logged up to 08:37 on 8 Oct. Their status:

- **Fixed in code or procedure:** twenty-eight. Rows 30 and 31 were expected or harmless; row 33 has a procedural fix, and its code fix is still open.
- **Documented hardware limits:** four (the full-motor attempts, the three-phase rotor model, and the two 1.0 mm-mesh Rev1 checks, rows 36–37).
- **Cosmetic:** one.
- **Root cause not determined:** two, the K1 freeze and the K1 crash (rows 25 and 32).
- **A methodology finding:** one, the region-boundary bias exposed by the validation gate (row 34).
- **Not in the field:** one, an Octave limit that affected only Claude's cross-check.

The fixes are part of the method: they explain why the pipeline is stepwise, checkpointed, restartable and gated.

#	When	Symptom (exact message where known)	Root cause	Fix (version)
1	3 Oct	"Error in calculating the conduction paths. The most likely reason is that the mesh of the source conductor is wrong."	Internal "loop" terminals in Transient	Boundary terminals: raised leads to the region +Y face, padding 0
2	4 Oct 18:56 (review)	Gate 2 created an ordinary Transient, not A-Phi	One solution-type string used for both formulations	Separate formulation key; native A-Phi setup and report type (v2–v3)
3	4 Oct (review)	Resume could duplicate partial objects; "OK" without proof	Saved after a failed stage; no acceptance gate	Checkpoint after each successful stage; reject a solve with error messages or no solution (v2–v3)
4	4 Oct 20:43 / 22:18	Gate 2 solved with 0 A; the long form ran out of RAM after step 1	Design variables created after the windings; RAM	Variables first (v5); short form 2 steps, 5 mm padding
5	4 Oct 23:23	"The file Rev0.asol_priv is in use - do you want to wait?"	Reused project name, lock files	Never overwrite an existing project (v5.1)
6	5 Oct 00:26 and 03:02	PC freeze at "Start Matrix" (6.2 GB T-Omega; 8.04 GB MS)	Inductance-matrix step on 16 GB	Matrix off by default (v5.2); `postprocess_ms.py` recovers the last pass
7	5 Oct 03:30	"calculator step failed (EnterQty(string quantityName) (Ohmic_Loss) - call failed"	Field quantity name	Fall back to "OhmicLoss"
8	5 Oct 16:33	"Can not create property named 'theta' because it conflicts with an existing predefined, intrinsic, or internal variable."	"theta" is reserved in AEDT	Variable renamed `rot_angle`
9	5 Oct about 16:45	"Analyze(…) -> long(ParametricSetup1) - call failed"	Optimetrics setup analysed with the wrong call	Optimetrics solved through its own call (exact change not recorded)
10	5 Oct 22:21	Lab PC crashed mid-baseline; nothing saved	Optimetrics saves only at the end	Stepwise sweep: save and `sweep_progress.json` after every solve; resume at "analyze"; reopen the project between angles (v6.6)

#	When	Symptom (exact message where known)	Root cause	Fix (version)
11	6 Oct 02:08	"Loading project failed: REV0_REF_ROTOR_MS_SWEEP."	Reopen after the crash	Restore the validate checkpoint
12	6 Oct (more than 2.5 h in)	v7 build: AEDT closed	One united solid per phase across 5 boards	No cross-board union, purge history per board, per-board checkpoints, stator-only build (v7.1–v7.3)
13	6 Oct 20:54	"PCB1 failed the AEDT Validation Check" ("No excitation…", "No solution setups…")	Build-only notices treated as errors	Accept the expected build-only notices (v7.4)
14	7 Oct 00:15	"Illegal external terminal 'PhaseC_In': An external terminal must border the edge of the problem region…" (all 6)	Region +Y padding 5 mm	Hand fix; v7.5 forces +Y padding 0 when leads exist
15	7 Oct	"Baseline summary has no fit_coeffs"	Baseline exported by v6	`fix_baseline_summary.py` re-analyses without re-solving
16	7 Oct about 17:30	Still "no fit_coeffs"	IronPython 2.7 `ast.literal_eval` rejects negative numbers	Fallback parser `_literal()` (v9.2)
17	7 Oct about 17:30	UnboundLocalException 'cfg' in `_publish()`	Masked the real error: an old config.py on PC2	Initialise `cfg`; startup guard "config.py too old" (v9.2)
18	7 Oct	`stack_sum.py` gave K_S1 = 40	Closure bug: loop variable reused	Bind by default argument
19	7 Oct	"Can not create property named 'rot_angle' / 'lamp' / 'I_A'…"	Variable already in the copied project	Harmless: falls back to changing the property (cosmetic, open)
20	7 Oct about 18:14	RedLab17 crash: 95 % memory, solver 8.64 GB, pass 1	Three phases + rotor on one board too heavy	Replaced by B/C superposition
21	7 Oct 18:08, about 19:00, 23:02	Full five-board attempts froze	16 GB RAM (see "Full-motor solves and the computational limit")	Documented limit; no fourth attempt
22	7 Oct	Attempt-1 status file overwritten	Local folder wiped on restart; numbering reset	Numbering also checks the shared folder (v9.3)
23	7 Oct about 22:30	`phase_check.py` sign error	−PHASE_DEG instead of +PHASE_DEG	Fixed before delivery
24	7 Oct 22:56	Rev1 spec conflict: S2 faced bare steel	Specification, not code	Rev1-D added; Rev1-S kept (v9.4)
25	7 Oct 23:44	REV1D_S2_K1 froze about 21 min in	Not determined; the same model crashed again at 02:02 on 8 Oct (row 32)	Clean restart, rerun from 00:28
26	8 Oct, before 01:36 (offline)	Trace DC network: singular matrix	Disconnected pieces: chords not joined at pads and vias	Group chords by copper island, split segments at vias, join nodes inside each pad (radius + 0.26 mm), add a 1e-9 leak (v9.5 trace build)
27	8 Oct, before 01:36 (offline)	Terminal node not attached to the winding network	Terminal coordinate not on a track end	Attach within 2.0 mm and join every node on the terminal pad; final R_dc 0.2656 Ω
28	8 Oct, before 01:36 (offline)	Analytic check gave |B| 0.076 T and K1 1.63 mN·m/A	Back-plate image with the wrong sign	Ideal-iron image of opposite magnetic charge: K1 6.110 mN·m/A

#	When	Symptom (exact message where known)	Root cause	Fix (version)
29	8 Oct (offline)	Analytic field model ran out of memory or timed out	Too many field points per evaluation	Smaller chunks, 0.5 mm integration step, 4 angles, background run
30	8 Oct, about 02:10	"Field export variant 0: echoed points do not match the requested ones"	ExportToFile reads the points in SI units unless the file states otherwise	Expected: the 20-point probe chose variant 1 ("Unit=mm" header, PtInSI False) automatically (v9.5)
31	8 Oct, about 01:40	K3 rotor_field_metrics.csv blank at 11.25°	Field export failed at that one angle	Not needed: three angles give the plate field and force; K1 is from the torque, which was complete
32	8 Oct, 02:02	RedLab17 crashed during REV1D_S2_K1's fourth angle	Not determined (second failure of this two-rotor model)	Restart; the run finished at about 04:00
33	8 Oct, found 02:49	K1 started again from angle 1 after the restart	The lab PC wipes C:\Users\Student\Rev0_work on restart, and the step progress lived only there	Procedure: after a crash, close AEDT without logging out. Saving step progress to OneDrive proposed, not yet implemented
34	8 Oct, 03:05	"VALIDATION GATE … FAIL (K1 +4.12 %, psi −0.02 deg; limits 2 % / 1 deg)"	The detailed reference read low: its B·n = 0 region face sat 5 mm below the board. Mesh ruled out (0.03 % and 0.01 % changes)	Region padding 20 mm: reference 6.2816, gate +1.40 %, PASS at 05:50 (padding runners)
35	7–8 Oct (Claude's cross-check)	GNU Octave: ode15s fails on Model 2; no graphics	Octave's IDA-based ode15s and the container have no plotting	Model 2 checked with a Python mirror (m2_rev1.py); plotting stubbed. MATLAB itself unaffected
36	8 Oct, about 07:00	RedLab18 froze in REV1D_S2_K3_M10	RAM: 1.0 mm mesh in both 3 mm gaps of the two-rotor model	Documented limit; fallback runner with the original mesh offered; check stopped
37	8 Oct, 08:19	RedLab17 froze in REV1D_S2_K3_P20_M10	As 36, with a larger air region	As 36

Engineering decisions

Olwethu made the design decisions, or adopted them from the reviewer. Claude's proposals only became method once one of them accepted them. Two items still need Olwethu's confirmation (the rows marked "Awaiting Olwethu"). Rows from 8 Oct are at the end.

When	Decision	Rationale	Decided by
4 Oct 05:02	Specification authoritative; Claude executes, flags conflicts, never changes the spec	Clear division of responsibility	Olwethu

When	Decision	Rationale	Decided by
4 Oct 05:02	Terminals L1 In / L6 Out; 615 vias; 19 mm shaft; 5 boards in series, same orientation	Specification	Olwethu with the reviewer
4 Oct 18:56	T-Omega for any rotating model; A-Phi only for stationary checks	A-Phi has no motion in AEDT 2025 R2	Reviewer
4 Oct 18:56	Shaft modelled on the rotor side only and united with the back plate	Electromagnetic simplification; robust geometry	Reviewer, on Claude's question
5 Oct	FR-4 and solder mask modelled as air in magnetostatic runs	μr = 1 and non-conductive; checked: ⩽ 0.22 % change	Olwethu
5 Oct	Inductance matrix off by default	Two PC freezes at "Start Matrix"	Claude, accepted
5 Oct 05:06	Single board validated; Gate 2B deferred; terminal architecture frozen	Gate evidence	Reviewer and Olwethu
5 Oct	Build first, verify, freeze; test only on copies of frozen builds	Robustness	Olwethu
5 Oct 16:26	Torque reciprocity method with a full-period baseline; gap mesh ⩽ g/2	Method rigour	Reviewer
5 Oct 18:30	Locked order: baseline → three-phase check → DOE → board-by-board stack	Validation chain	Reviewer
5–6 Oct	Single-current runs use the copper (Lorentz) torque	Rotor virtual-torque bias of about −0.47 mN·m	Claude, accepted

When	Decision	Rationale	Decided by
6 Oct 16:57	Bridge board k L6 → board k+1 L1; star point not made	Polarity not yet proven	Reviewer and Olwethu
6 Oct	Stator built board by board, stator only, rotor added later	v7 crash	Olwethu
7 Oct 02:30–03:35	Reduced DOE (0.5 mm and 5.0 mm gaps, 5 mm plate) with an exponential distance law, instead of a 12 h full DOE	16 GB limit; coreless field superposes	Claude, accepted by the reviewer and Olwethu
7 Oct 13:03	First Rev1 spec: S1–R1–S2–R2, same g everywhere, rotors aligned, stators in series	Deadline	Reviewer, adopted by Olwethu
7 Oct	Shared OneDrive folder; solve on local disk	One flash drive, several PCs	Olwethu
7 Oct 16:15	**Rev0 locked: g = 1.0 mm, t = 2.0 mm**	5 mm plate gives +0.1 % for about +130 g; 1 mm leaves clearance for PCB bow, runout and tolerances	**Olwethu**
7 Oct 17:23	PC1 runs the reduced method; PC2 deliberately attempts the full solve and documents the failure	Evidence for the conclusions	Reviewer, accepted by Olwethu
7 Oct, evening	Fewer passes and a coarser mesh where possible; results first	Time	Olwethu
7 Oct, evening	Three-phase gate by single-phase superposition (B and C solved separately)	Three-phase model crashed; copper torque linear to 0.005 %	Claude, accepted

When	Decision	Rationale	Decided by
7 Oct about 21:20	Build a FEM-informed MATLAB lumped model, as a complement	System-level dynamics	Olwethu; reviewer endorsed
7 Oct 22:30–22:41	Rev0 validated and closed at three levels; full five-board solve not required	All gates passed	Reviewer and Olwethu
7 Oct 22:56	Rev1-D (48 magnets) primary; Rev1-S comparator; R1 plate thickness a parameter; S2 polarity not hardwired	S2 faced bare steel in the old Rev1	Reviewer, forwarded by Olwethu
7 Oct 23:13	No fourth full-solve attempt	Three failures on two PCs	Claude recommendation
7 Oct 23:05	"Through" magnetisation for Rev1-D	Fields add across S2; R1 plate lightly loaded	**Awaiting Olwethu**
7 Oct 23:05	Test Rev0 hardware as S1 + R2 (48 magnets cover all builds)	60 magnets available	**Awaiting Olwethu**
8 Oct 00:40	Motor construction complete; no further meshed-copper full-motor solves on 16 GB	Three failed attempts; RAM analysis (mesh, direct-solver matrix, adaptive passes)	Claude's assessment, accepted
8 Oct 01:10–01:36	Full-motor model by field-only FEM with trace-resolved winding post-processing (trace centre-lines as line currents), not equivalent conductor blocks	Keeps the real winding path; nothing to calibrate; no copper mesh, so it fits in 16 GB	Olwethu ("Essentially option 1"); reviewer endorsed

When	Decision	Rationale	Decided by
8 Oct, before 01:36	Board 1 must reproduce the detailed K1 within 2 % (and ψ within 1°) before any field-map number is used; never call the method "exact"	Validation before use	Reviewer
8 Oct 01:41	S2 wired in series in the same sense as S1 (S2_REVERSED = False) in both Rev1 designs	S2 vs S1 phase +0.1° (K3), then 0.0° in all four Rev1 field maps	FEM evidence; Claude recommendation
8 Oct 02:10	Rev1-D shared R1 plate kept at 2 mm	|B|max 1.23–1.30 T, below saturation	FEM evidence; Claude recommendation
8 Oct 03:05	Field-map numbers withheld while the gate failed; mesh convergence checked first	Gate rule	Claude, under the reviewer's rule
8 Oct 04:24	Test the region boundary: 20 mm padding for the single-board reference and the three field maps	Analytic B·n = 0 model predicted the 4 % gap	Claude, accepted by Olwethu
8 Oct 04:27	RedLab19 (PC3) moved to the thermal model	Thermal workstream	Olwethu
8 Oct 04:33	Close the gate with two PCs (PC1 reference, PC2 field maps); Rev1 padding runs added on PC2	Fewest runs to settle the absolute level	Olwethu, on Claude's plan
8 Oct 05:50	Gate passed: field-map values (20 mm padding) are the headline results; board-by-board values are the reduced-method cross-check; MATLAB uses USE_FIELDMAP = true	+1.40 %, 0.04° against 2 % / 1°	Reviewer's pre-agreed rule, applied by Claude

When	Decision	Rationale	Decided by
8 Oct 05:52	Electromagnetic modelling complete; thermal modelling continues separately	All gates passed; all designs solved	Olwethu
8 Oct 08:37	Stop the electromagnetic runs; report the Rev1 S2 discrepancy as an open point and a limitation	Both check runs froze; no conclusion depends on it; PCs needed for the thermal work	Olwethu

Rev0 validated results

This section records the board-by-board (reduced) method as it stood on 7 Oct. At g = 1.0 mm and t = 2.0 mm it gave K_stack = 18.05 mN·m/A per phase, 27.1 mN·m per A_pk, 1.336 V rms at 1000 rpm and Km = 0.0188 N·m/√W. Every link in the chain was checked separately, but the absolute level is about 8 % low. The single-board reference runs had the region boundary only 5 mm below the board. The final, boundary-free value is 19.50 mN·m/A (section "Final electromagnetic results"). The ratios, phases, three-phase check and quality checks below are unaffected, because they compare models with the same boundary.

Method. One detailed board with the real rotor is solved magnetostatically at a series of rotor angles. The torque on the copper gives the flux-linkage derivative, which is both the torque constant and the back-EMF shape:

$$k(\theta) = \frac{d\lambda}{d\theta} = \frac{T(+I) - T(-I)}{2I}, \qquad e = \omega\,k(\theta), \qquad T = \sum_x i_x\,k_x(\theta)$$

The stator has no iron, so each board's contribution is independent of the others. Board k sits at distance d_k = g + (k−1)p from the magnets, with p = 1.02032 mm, and the five-board string adds the boards as phasors:

$$K_{stack}(g) = \left|\sum_{k=1}^{5} K_1(d_k)\,e^{j\psi(d_k)}\right|, \qquad K_1(d) = K_1(1\,\text{mm})\,e^{-0.3031\,(d-1)}$$

Single board, Phase A, at 1.0 mm [AEDT]

Quantity	Value
Resistance per board-phase (A / B / C)	0.27776 / 0.27736 / 0.27789 Ω (mean 0.2777, +2.9 % on 0.2699)
Self inductance per board-phase	about 2.07 μH
K1 (= dλ/dθ fundamental peak)	6.0917 mN·m/A, phase ψ = −30.18° electrical
PM flux linkage, back-EMF	0.7615 mWb peak; 0.451 V rms at 1000 rpm
3rd harmonic	H3/H1 = 11.75 %
Quality checks	Repeat 0.019 %; antisymmetry 0.051 %; rotor vs copper torque ⩽ 0.116 %; nonlinearity 0.005 %
Field	Steel

Distance law and back plate [AEDT, fit in Py]

Run	Rotor distance	Plate	K1 (mN·m/A)	ψ (°)	Fit residual
REV0_DOE_D050	0.5 mm	2 mm	7.11	−30.4	−0.15 %
Baseline	1.0 mm	2 mm	6.0917	−30.18	+0.17 %
REV0_DOE_D500	5.0 mm	2 mm	1.815	−30.0	−0.02 %
REV0_DOE_BI50	1.0 mm	5 mm	6.097 (+0.1 %)	−30.18	—

- The fitted decay rate is k = 0.3031 /mm, close to the analytic fundamental π/τ = 0.2857 /mm (pole pitch τ = 11.0 mm at r = 28 mm).
- The 3rd harmonic decays faster: k3 = 0.7386 /mm.
- The phase hardly moves with distance (−30.36 + 0.071·d degrees), so the five boards add almost in phase.
- Stack factors at g = 1 mm: S1 = 2.958 for the fundamental and S3 = 1.846 for the 3rd harmonic.
- Board shares of the stack: 34 / 25 / 18 / 13 / 10 %. Board 5 sees about 30 % of board 1's field.

Boundary note (found 8 Oct). The decay rate of 0.3031 /mm is steeper than the true one. The single-board points were suppressed by the region boundary, and more so the farther the board was from the magnets. The full-motor model with 20 mm padding gives 0.282 /mm through the stack, close to π/τ = 0.286 /mm. The boundary-free board shares are 33 / 25 / 18 / 14 / 10 %, which is very close to the reduced method's.

Junction and stack [AEDT]

- **Two-board junction test:** 0.55313 Ω against 0.5554 Ω expected (−0.4 %). The bridge loss is 3.6 µW, so both boards carry the full current. The string inductance is about 6.23 µH, giving an adjacent-board mutual of about 1.04 µH [Est].
- **Phase resistance:** R_φ = 5 × 0.2777 = 1.389 Ω, excluding external leads.
- **Phase inductance of the five-board string:** 18.7–31.1 µH, nominal 24.9 µH [Est].

Three-phase check by superposition [AEDT + Py] — PASS

Phase	K1 (mN·m/A)	vs A	ψ (°)	Expected ψ	Error
A	6.0917	—	−30.18	—	—
B	6.1107	+0.31 %	+90.02	+89.82	+0.20°
C	6.0935	+0.03 %	−149.78	−150.18	+0.40°

- **Mean synchronous torque:** 9.1478 mN·m per A_pk per board, against the ideal 1.5·K1·I = 9.1375 (+0.11 %).
- **Reversed sequence (A-C-B)** would give −0.0144 mN·m. So the phase order and polarity are proven.
- **Tolerances:** 2 % on K, 2° on ψ, 2 % on torque.

Rev0 stack at g = 1.0 mm, t = 2.0 mm [Py on AEDT]: K_stack = 18.05 mN·m/A per phase (2.96 × one board); T = 27.1 mN·m per A_pk; E1 = 1.336 V rms at 1000 rpm; Km = 0.0188 N·m/√W.

The final Rev0 model `REV0_FINAL_G100_T20_FROZEN.aedt` passed all 36 junction probes, with the gap at 1.0000 mm. Its axial levels are: stator −4.59144 to 0.51016 mm, magnet face 1.51016 mm, back plate top 5.51016 mm.

Full-motor solves and the computational limit

The complete detailed Rev0 motor (five boards × three phases + rotor) could not be solved on a 16 GB lab PC. Three attempts on two PCs failed without one solved field, during meshing or during the first pass. The geometry itself is valid: every final model passed AEDT's validation check and all junction probes. On 8 Oct the full motors were solved a different way, without meshing the copper (next section).

The three attempts (copy of REV0_FINAL; 45 solids: 27 copper, 16 magnets, 1 steel body; one rotor angle; synchronous currents I_A +0.864, I_B +0.003, I_C −0.868 A; magnetostatic, 2 passes, matrix off)

Attempt	PC	Timeline	How it ended
1	RedLab18	Froze 18:08	Nothing recorded (status file later overwritten)
2	RedLab18	18:30 opening at 55 % memory → 18:59 still in "Making Initial Mesh … Fixing local region mesh issues (2)"	Froze at about 19:00 in initial meshing. Mesher alone 5.5 GB, 62 % memory
3	RedLab19	Start 21:13:00; open 154 s; validation check about 9 min; solve start 21:24:59 with 4.03 of 15.81 GB free	After 1 h 37 min, no adaptive pass finished. AEDT closed and the PC froze at 23:02

What consumes the memory

1. **The mesh.** Maxwell fills the model with tetrahedra, and an element cannot be much larger than the thinnest feature next to it. The copper is 35 µm thick, and adjacent boards are only 20 µm apart, mask to mask. So every trace layer needs a very fine mesh, repeated over 6 layers × 3 phases × 5 boards (× 2 stators for Rev1). Attempt 2's message "Fixing local region mesh issues" is the mesher struggling with exactly those thin gaps.
2. **The matrix.** Every element adds unknowns. Maxwell's default direct solver factorises the whole matrix, and its memory grows faster than the number of unknowns.
3. **Adaptive passes.** Each pass refines about 20 % of the elements, so memory peaks on the last pass.

Measured scale points [AEDT]

Model	Phase-boards	Peak observed	Outcome
One board, one phase, + rotor	1	Solves in 11–21 min per angle	Solves
One board, three phases, no rotor, no matrix	3	—	Solves
One board, three phases + rotor	3	Solver 8.64 GB at the start of pass 1, 95 % memory	Crash
Full Rev0 motor	15 + 12 bridges	11.8 GB in use before solving; mesher 5.5 GB	Freeze

Model	Phase-boards	Peak observed	Outcome
Full Rev1 motor	30 + 24 bridges	Not attempted	—

Extrapolation to bigger machines [Est — order of magnitude only, nothing measured]

The full Rev0 has 5 times the copper of the three-phase single-board model, which already needed 8.6 GB in its solver at pass 1. Rev1 has 10 times. Assuming the default direct solver, 2–4 passes, and one rotor angle:

RAM	Full Rev0	Full Rev1-S or Rev1-D
32 GB	Will not solve	Will not solve
64 GB	Borderline; perhaps with 2 passes or the iterative solver	Will not solve
128 GB	Should solve; roughly 4–24 h per angle, depending on cores	Borderline; iterative solver, 256 GB or HPC

- A rotor-angle sweep needs 4–9 angles, so days for Rev0.
- A transient with rotation needs about 90 time steps per electrical period, so weeks. That is not realistic at these sizes.
- The run times depend strongly on the CPU; the lab PCs′ core count was not recorded.

Why building the model by hand would not help. A hand-built model is the same model: the same mesh, the same matrix, the same memory. The script only made the build repeatable and checked every junction (72 junction probes across the two stators of Rev1-D).

Why the reduced method is valid. The board-by-board method is not a workaround. With a coreless stator, linear magnets and unsaturated steel, each board′s flux linkage adds independently. This was confirmed in three ways:

- the copper torque is linear to 0.005 %;
- the three-phase superposition reproduces the ideal torque to 0.11 %;
- the distance law fits three solved distances to within 0.17 %.

Recommendations for anyone who needs a full-motor cross-check:

1. Switch Maxwell to the iterative solver. It typically needs several times less memory, at the cost of longer and less certain convergence.

2. Replace the 35 µm traces with equivalent conductor blocks carrying the same current distribution. This is the standard way to model PCB windings and cuts the mesh drastically. On 8 Oct a stronger version was used instead: the copper was left out of Maxwell entirely, and the field was integrated along the real trace paths (next section).
3. Use a periodic sector model if the winding repeats every few poles. This needs checking against the layout first.
4. Run one full-model check on a workstation with ≥ 128 GB, or on the UCT HPC cluster.

Full-motor field-only FEM and the validation gate

On 8 Oct the full Rev0, Rev1-S and Rev1-D motors were solved on the 16 GB lab PCs in 7–25 min each. Maxwell solved the magnet field without the copper, and Python integrated the Lorentz force along the real PCB current paths. The first gate failed (+4.1 %). The cause was the region boundary in the reference runs, and after correcting it the method passed at +1.40 % and 0.04°.

Why leaving the copper out is valid.

- The stator is coreless. Copper and FR-4 have µr ≈ 1, and in a magnetostatic magnet-field solve they carry no current, so removing them does not change B.
- By reciprocity, the torque per ampere that the magnet field exerts on the conductors is k(θ) = dλ/dθ.
- Two checks already supported this: modelling FR-4 as air changed results by ⩽ 0.22 %, and the copper torque was linear to 0.005 %.

Torque on the winding. The sum runs over every track piece and via step i of a phase, on every board and both stators, with I_i the DC current in piece i:

```
T_z = \sum_i I_i \left[\, dl_{z,i}\,(x_i B_{x,i} + y_i B_{y,i}) \;-\;
B_{z,i}\,(x_i\, dl_{x,i} + y_i\, dl_{y,i}) \right], \qquad k_x(\theta) = -
\frac{T_{z,x}(\theta)}{I_x}, \qquad e_x = \omega\, k_x(\theta)
```

For a flat track dl_z = 0, so only the B_z term remains. For a via only dl_z ≠ 0. K is the fundamental amplitude of a Fourier fit (harmonics 1 and 3) over the 8 angles.

Item	Setting
Field solve	Magnetostatic, 4 adaptive passes, 20 % refinement, inductance matrix off
Bodies	16 / 32 / 48 N35 magnets; steel_1010 rotors united with the 19 mm shaft; air. No copper, no FR-4

Item	Setting
Mesh	Air zone around each stator (r 13.5–47.5 mm) at ⩽ 0.8 mm, at most 500,000 elements; magnets 2.0 mm
Region padding	5 mm in the first runs, 20 mm in the final runs
Rotor angles	8, from 0 to 39.375° mechanical in 5.625° steps: one electrical period (8 pole pairs)
Winding paths	ODB++, six layers; 0.4 mm chords; about 11,040 track pieces and 1,015 via steps per board-phase. A DC nodal solve gives each piece's current, so the parallel layers L2/L3 and L4/L5 share correctly. R_dc 0.2656 Ω
Sample points	180,775 (Rev0); 361,550 (Rev1)
Field export	FieldsReporter.ExportToFile with a points file; variant 1 ("Unit=mm" header) chosen by a 20-point probe
Time per rotor angle	0.8 min (Rev0), 1.5 min (Rev1). One meshed-copper board takes 11–27 min
Neglected	Armature reaction; eddy, skin and proximity effects in the copper; averaging of B over the 0.5 mm × 35 µm cross-section (B is sampled on the centre-line)
Resistance used	The meshed-copper FEM value, 0.2777 Ω per board-phase

Gate history. The gate compares the field map's board 1 with a detailed single board that has meshed copper, at g = 1 mm.

Time (8 Oct)	Detailed reference K1	Field-map board 1 K1	Difference	Verdict
03:05	6.0917 (5 mm padding)	6.3429 (5 mm padding, 0.8 mm mesh)	+4.12 %, −0.02°	FAIL
04:01	6.0917	6.3408 (0.55 mm mesh)	+4.09 %	FAIL: mesh ruled out
04:45	6.0917	6.3697 (20 mm padding)	+4.56 %	FAIL against the old reference

Time (8 Oct)	Detailed reference K1	Field-map board 1 K1	Difference	Verdict
05:50	6.2816 (20 mm padding)	6.3697 (20 mm padding)	+1.40 %, 0.04°	PASS

The boundary mechanism. Maxwell's default outer boundary makes B tangential to the region faces (B·n = 0). For a field source this acts like a same-sign mirror image across the face, which flattens B_z between the face and the board.

- The hand-calculated field models each magnet face as a sheet of magnetic charge. The steel plate is modelled as an ideal-iron image of opposite sign, and the boundary as a same-sign image.
- With this model, a face 5 mm below a single board lowers K1 by 4.1 %. A face 5 mm below the whole stack lowers board 1 by only 0.35 %. That predicts a field-map/reference ratio of 1.039; the observed ratio was 1.041.

Convergence and consistency checks.

Check	Change	Reading
Field-map zone mesh 0.8 → 0.55 mm	Board 1 −0.03 %	Mesh converged
Detailed gap mesh 2.5 → 1.0 mm (5 mm point)	−0.01 %	Mesh converged
Field-map padding 5 → 20 mm, Rev0	Board 1 +0.42 %, board 5 +3.9 %, stack +1.2 %	Boundary effect, as predicted (+0.35 %, +3.6 %)
Detailed padding 5 → 20 mm	+3.1 %	Boundary effect on the reference
Field-map padding 5 → 20 mm, S2 stacks	+0.03 % (Rev1-D), +0.01 % (Rev1-S)	S2 is shielded by steel on both sides, as predicted
Phases A / B / C	Within 0.25 % in every design	Balanced
Field map vs corrected reference	+1.40 %	Within the 2 % gate. Likely cause: centre-line sampling

Per-board results, 20 mm padding (K1 in mN·m/A, phase A). S1 board 1 faces R1. S2 board 1 faces R2, and S2 board 5 faces R1.

Board	Rev0 S1	Rev1-D S1	Rev1-D S2	Rev1-S S1	Rev1-S S2
1	6.3697	6.3803	8.7587	6.3713	6.6506
2	4.7962	4.8016	7.8497	4.8006	5.1866
3	3.5881	3.5937	7.5412	3.5886	4.0853
4	2.7268	2.7251	7.8575	2.7244	3.3286
5	2.0154	2.0222	8.7539	2.0159	2.8842
Stack K	19.496	19.523	40.761	19.501	22.135
Stack H3/H1	5.9 %	5.9 %	5.7 %	6.0 %	5.3 %

- Every board phase lies between −30.21° and −29.94°, so the boards add in phase.
- **Caveat for Rev1 (not yet resolved).** The gate validated a single-sided S1 board only. The field-map S2 boards read above the detailed Rev1 single-board runs, which used 5 mm padding:
  - Rev1-D board next to R1 (K1 run): 8.7539 against 8.4860, +3.2 %.
  - Rev1-D middle board (K3 run): 7.5412 against 7.0504, +7.0 %.
  - Rev1-S board next to R1's steel (S2_ENV run): 2.8842 against 2.5970, +11 %.
  - The padding runs show S2 is not sensitive to the outer boundary, so that boundary is not the explanation here. The cause has not been established.
  - The design ratios agree within 1–2 % between the two methods. The Rev1 absolute values carry this open uncertainty; a REV1D_S2_K3 rerun with 20 mm padding would test it (Open items).
- The "action/reaction mismatch" lines in the detailed summaries (14–89 %) compare against Maxwell's rotor-side torque. That value is unreliable with +I only and large gaps. K is always taken from the copper torque.

Rev1: two designs

Rev1 adds a second stator. It became two designs on 7 Oct at 22:56. Rev1-D (48 magnets) is the primary design; Rev1-S (32 magnets) is the comparator. Both final geometries can be built with the same pipeline, and Rev1-D is built and frozen. Final values from the full-motor model: Rev1-D K = 60.28 mN·m/A (3.09 × Rev0) and Rev1-S K = 41.64 mN·m/A (2.14 × Rev0). In both designs S2 is wired in the same sense as S1.

Label	Topology	Magnets	Role
Rev0	S – R	16	Validated baseline

Label	Topology	Magnets	Role
Rev1-D	S1 – [R1: 16 PM | shared steel | 16 PM] – S2 – [R2: 16 PM | steel]	**48**	**Primary**
Rev1-S	S1 – [R1: 16 PM | steel] – S2 – [R2: 16 PM | steel]	32	Comparator

Common to both designs:

- g = 1.0 mm on every active gap;
- R2 steel 2 mm;
- rotors aligned at 0° on one 19 mm shaft;
- two identical five-board stators in series per phase, so $R_\varphi \approx 2.777\ \Omega$.

The thickness of R1′s shared plate in Rev1-D is a parameter, starting at 2 mm, to be checked from the FEM |B|. The S2 series polarity was not hardwired; it was set from the FEM phase (result: same sense as S1, S2_REVERSED = False).

Magnetisation (Claude′s recommendation, awaiting Olwethu). At any angle, every magnet in the motor has the same axial magnetisation vector. This has three effects:

- R1′s face B and R2 then show opposite poles across S2, so their axial fields add.
- Flux passes straight through R1′s plate, which carries little circumferential flux. The alternative ("mirror") would double it.
- S1 and S2 see the field in the same direction, so they are expected in phase.

Method for S2 in Rev1-D. S2 board j (1 = next to R1, 5 = next to R2) sits at $d_L = g + (j-1)p$ from R1′s face B and $d_R = g + (5-j)p$ from R2. Superposition of the single-sided law predicts:

```
P_j = K(d_L)\,e^{j\psi(d_L)} + K(d_R)\,e^{j\psi(d_R)}, \qquad K_{S2} =
\Big|\sum_{j=1}^{5} f_j\,P_j\Big|
```

Two dedicated FEM boards check this prediction, each one board between two PM rotors at 4 angles:

- **REV1D_S2_K1:** the board next to R1 (gaps 1.0 mm and 5.08 mm).
- **REV1D_S2_K3:** the middle board (3.04 mm from each rotor).

Each gives a correction $f_j = |\text{FEM}| / |\text{prediction}|$. The other boards use mirror symmetry and interpolation. These runs also report the peak |B| in R1′s plate and in R2′s steel, and an estimate of the axial force pulling R2 towards R1.

Method for S2 in Rev1-S. Each S2 board sees R2, plus an image of R1′s bare steel across the stack, with factor $1 + c \cdot e^{(-2k(L-d-p/2))}$ where $L = 2g + 5p$. The coefficient c is calibrated on REV1_S2_ENV: the S2 board next to R1′s steel, with R2 at g + 4p.

Pre-FEM prediction, Rev1-D (history, superseded by the FEM results below) [Py, pure superposition, g = 1 mm]

- S2 per-board K: 7.87 / 6.89 / 6.57 / 6.89 / 7.87 mN·m/A.
- K_S2 = 36.10 and K_Rev1-D = 54.14 mN·m/A, which is 3.00 × Rev0.
- T = 81.2 mN·m per A_pk; E1 = 4.009 V rms at 1000 rpm; Km = 0.0398 N·m/√W (2.12 × Rev0).
- Expected K3 copper torque at θ = 0: about ±5.7 mN·m, against about 5.3 for a Rev0 board at 1 mm.

Rev1-S without the steel image is 2 × Rev0 (Km 1.41 ×). The image adds to that.

Final build REV1D_FINAL_G100_T20 [AEDT] — built, validated and frozen on RedLab19 (23:28 → about 00:17)

Level	z (mm)
S1 bottom / top	−4.591 / 0.510
R1 face A (magnets toward S1)	1.510
R1 plate top	5.510
R1 face B (magnets toward S2)	7.510
S2 bottom / top	8.510 / 13.612
R2 magnets / steel top	14.612 / 18.612

- S2 is a copy of S1 shifted 13.10160 mm along +Z. Junction probes passed: S1 36/36 and S2 36/36. The geometry check passed.
- 48 magnets turn with the `rot_angle` variable. S2 is wired in series in the same sense as S1, provisionally.
- **Build time 48.6 min:** each 32-magnet duplication took about 400 s, and the validation check about 19 min.

Board-by-board FEM results, 8 Oct [AEDT]. These runs used 5 mm padding and the g/2 gap-mesh rule. The final absolute values come from the full-motor model.

Run	Board position	Detailed K1 (mN·m/A)	Prediction before FEM	Factor	ψ	H3/H1	R1 plate peak B	R2 steel peak B	Axial force on R2
REV1D_S2_K1	Rev1-D S2 board next to R1 face B (1.0 mm; R2 at 5.081 mm)	8.4860	7.8711	f = 1.0781	−30.16°	8.93 %	1.27–1.30 T	1.41–1.49 T	−47.2 to −47.3 N
REV1D_S2_K3	Rev1-D middle S2 board (3.041 mm from each rotor)	7.0504	6.5731	f = 1.0726	−30.04°	4.49 %	1.23–1.25 T	1.42–1.47 T	−47.7 to −47.8 N
REV1_S2_ENV	Rev1-S S2 board next to R1′s bare steel (1.0 mm; R2 at 5.081 mm)	2.5970	1.7707 (no steel)	×1.467, so c = 1.166	−30.05°	1.51 %	—	—	—

- **Why f > 1.** Each rotor′s steel is a return path for the other rotor′s field, so the S2 boards gain 7–8 % over simple addition of the two single-sided fields.
- **Rev1-D stack sum.** The interpolated factors are 1.0781 / 1.0754 / 1.0726 / 1.0754 / 1.0781, giving K_S2 = 38.84 and Rev1-D = 56.89 mN·m/A (3.15 × Rev0).
- **Rev1-S stack sum.** With c = 1.166, K_S2 = 20.49 and Rev1-S = 38.54 mN·m/A (2.14 × Rev0).
- **Polarity.** S2 is in phase with S1 (+0.1°), so S2 is wired in the same sense.
- **Plate and steel.** The 2 mm shared R1 plate is unsaturated. It carries less flux than R2′s steel, as the "through" magnetisation predicts.
- **Mechanical load.** R1 and R2 attract each other with about 47–48 N (about 4.8 kgf), almost independent of angle. This is a Maxwell-stress estimate on the mid-gap surface. The shaft, spacers and bearings must hold both 1 mm gaps against it. The Rev1-S force was not computed.

Hardware note. 60 magnets are available. Rev1-D needs 48. Rev0 can be tested as S1 + R2, since R2 is identical to the Rev0 rotor, so all builds fit within 48 magnets.

MATLAB system model

The MATLAB model is the system level of a multiscale method. Detailed FEM of one board leads to a board-by-board reduction, which was then superseded by the full-motor field-only FEM. Those results feed a lumped electromechanical state-space model. The final run (MOTOR_MATLAB_MODEL_v2, 8 Oct 05:36) takes K and H3/H1 for each design from the gated full-motor model, and reproduces them exactly. Rev0 is final. For Rev1 the design ratios are final, while the absolute K carries the open S2 point described in the full-motor section.

Equations (Maxwell terminal convention; x = A, B, C; isolated-neutral star in Model 2):

```
e_x = \omega\,k_x(\theta), \quad T = \sum_x i_x\,k_x(\theta), \quad
J\dot\omega = T - T_L - B\omega, \quad L\,\dot i_x = v_x - v_n - R\,i_x -
e_x, \quad v_n = \tfrac{1}{3}\Big(\sum v - \sum e\Big)
```

Where each parameter comes from

Parameter	Value (Rev0)	Source
k(θ) shape	Baseline fundamental + 3rd harmonic (a1 5.26589e-3, b1 −3.06249e-3, a3 5.08187e-5, b3 −7.14189e-4 Wb/rad)	[AEDT]
Stack factors	S1 = 2.958 (k1 0.3032 /mm), S3 = 1.846 (k3 0.7386 /mm)	[Py on AEDT]
Phase B, C correction	Magnitude ×1.0031 / ×1.0003; angle +0.20° / +0.40°	[AEDT] three-phase check
R	1.389 Ω per phase	[AEDT]
L	24.9 µH (18.7–31.1)	[Est], sensitivity variable
J	1.13e-4 kg·m² (rotor 227 g, 50 mm shaft)	[Est], hand calculation
Drive	Vdc 12 V (SVPWM), I_limit 2 A_pk, B = 0	[Assump]
Speed loop (Model 1)	PI, Kp = J·20/Kt, Ki = Kp·20/2, anti-windup, current limit	[Assump]

The three models

- **Model 1, imposed currents.** Ideal current control, so L is not needed. Back-EMF, torque ripple (sine vs six-step), start-up and a load step.

- **Model 2, voltage-driven.** Solved with ode15s because the electrical time constant (about 18 µs) is far shorter than the mechanical one. Inductance sensitivity and six-step drive.
- **Model 3, steady state.** Current and speed sweeps, the torque–speed envelope at the bus voltage, and a copper-loss-limited efficiency map.

Rev0 results with the board-by-board K (MATLAB, 7 Oct 22:34, with the measured B/C phase data; absolute level superseded by v2):

- **Model 1:**
  - K_stack 18.02 mN·m/A; 27.03 mN·m per A_pk ideal.
  - At 1000 rpm: phase EMF 1.338 V rms, line-to-line 2.317 V rms.
  - At 1 A_pk: sine drive gives 27.06 mN·m with 0.70 % ripple; six-step gives 29.84 mN·m with 14.4 % ripple.
  - Start-up from 0 to 90 % of 1000 rpm takes 0.202 s.
  - At 20 mN·m and 1000 rpm: 0.737 A_pk, P_cu 1.123 W, copper-loss-limited efficiency 65.0 %.
- **Model 2** (2.5 V phase-peak sine supply):
  - 1306 rpm at no load and 787 rpm at 20 mN·m (59.5 %).
  - Identical for L = 18.7, 24.9 and 31.1 µH, because ωeL ≈ 0.016 Ω is about 1 % of R.
  - Six-step at duty 0.30: 1132 / 566 rpm, ripple 2.7 %.
- **Model 3:**
  - 12 V bus limits: 3671 rpm at no load; 3127 rpm at 20 mN·m.
  - Copper-loss-limited efficiency at 20 mN·m: 47.9 / 64.7 / 78.6 % at 500 / 1000 / 2000 rpm.
- **Agreement:** MATLAB's 18.02 mN·m/A and `stack_sum.py`'s 18.05 differ by 0.2 %, because the two fits use k = 0.3032 and 0.3031 /mm. Quote 18.05 as the electromagnetic result.

Rev1 extension (MOTOR_MATLAB_MODEL, 8 Oct). The Rev1 machine function is $k_x(\theta) = k_{x,S1}(\theta) + k_{x,S2}(\theta)$. The stack factors of S2 come from the same distance law:

- Rev1-D: two rotors per S2 board, × FEM factor f.
- Rev1-S: one rotor plus the steel image, × coefficient c.

R and L double for two stators in series. J and mass come from the geometry: Rev1-S 2.22e-4 kg·m² and 368 g; Rev1-D 2.46e-4 kg·m² and 401 g, with shaft extensions of 11.1 and 13.1 mm [Assump]. The scripts were cross-checked in GNU Octave (Models 1 and 3 reproduce the Rev0 MATLAB values exactly) and in the Python mirror (Model 2).

Final design comparison (MOTOR_MATLAB_MODEL_v2, run_all at 05:36:16 on 8 Oct; 12 V bus, 2 A_pk limit, load 20 mN·m at 1000 rpm):

Design	Magnets	K (mN·m/A)	T per A_pk (mN·m)	T per magnet	R (Ω)	Km (N·m/√W)	E at 1000 rpm (V rms)	Mass	No-load speed	Speed at 20 mN·m	η_Cu at 20 mN·m, 1000 rpm	0→1000 rpm
Rev0	16	19.50	29.2	1.828	1.389	0.0203	1.444	227 g	3393 rpm	2928 rpm	68.3 %	0.202 s
Rev1-S	32	41.64	62.5	1.952	2.777	0.0306	3.083	368 g	1589 rpm	1385 rpm	83.1 %	0.186 s
Rev1-D	48	60.28	90.4	1.884	2.777	0.0443	4.464	401 g	1097 rpm	1000 rpm	91.1 %	0.142 s

Kt is the ideal three-phase torque per A_pk. Km = Kt/√(1.5 R). E is the phase back-EMF in rms. No-load speed is the voltage-limited value. η_Cu counts copper loss only. 0→1000 rpm is at the current limit, no load, with ideal current control.

How v2 uses the full-motor values. motor_params.m sets USE_FIELDMAP = true. It holds K_fieldmap (Rev0 19.496e-3, Rev1-S 41.636e-3, Rev1-D 60.284e-3 N·m/A) and H3_fieldmap (5.9 / 5.6 / 5.8 %). The fundamental and 3rd-harmonic stack factors are rescaled so that each machine's K and H3/H1 equal the field-map values, and the k(θ) shape stays the FEM one. P.K1_boardsum keeps the board-by-board value for comparison. The FEM factors c_image = 1.166, fD_K1 = 1.0781 and fD_K3 = 1.0726 are built in. Setting USE_FIELDMAP = false reverts to the board-by-board values.

MATLAB run history (Olwethu's runs; K in mN·m/A for Rev0 / Rev1-S / Rev1-D):

Run	When	Rev1 inputs	K
MOTOR_MATLAB_MODEL, pre-FEM	8 Oct 01:33	Pure superposition; Rev1-S without the steel image (lower bound)	18.02 / 36.04 / 54.06
Same, FEM-calibrated	8 Oct 01:48	c = 1.166; fD_K1 = fD_K3 = 1.0726 (K1 not yet finished)	18.02 / 38.48 / 56.68
MOTOR_MATLAB_MODEL_v2 (final)	8 Oct 05:36	Full-motor field map, 20 mm padding; fD_K1 = 1.0781	19.50 / 41.64 / 60.28

The full-motor K is +8.2 %, +8.2 % and +6.1 % above MATLAB's board-by-board values (18.02 / 38.48 / 56.80).

First design comparison, pre-FEM (history only; superseded by the final table above) (12 V bus, 2 A_pk limit, 20 mN·m load):

Design	Magnets	K (mN·m/A)	T per A_pk (mN·m)	T per magnet	R (Ω)	Km (N·m/√W)	E at 1000 rpm (V rms)	Rotor mass	No-load speed	Speed at 20 mN·m	η_Cu at 20 mN·m, 1000 rpm	0→1000 rpm
Rev0	16	18.02	27.0	1.69	1.389	0.0187	1.33	227 g	3671 rpm	3127 rpm	64.7 %	0.219 s
Rev1-S (lower bound)	32	36.04	54.1	1.69	2.777	0.0265	2.67	368 g	1836 rpm	1563 rpm	78.6 %	0.215 s
Rev1-D	48	54.06	81.1	1.69	2.777	0.0397	4.00	401 g	1224 rpm	1103 rpm	89.2 %	0.159 s

Further results:

- **Model 2 at the same 2.5 V phase-peak supply** (no-load / loaded rpm): Rev0 1306/787, Rev1-S 654/393, Rev1-D 442/321.
- **Model 1 start-up to 90 % of 1000 rpm** under the speed loop: Rev0 0.202 s, Rev1-S 0.199 s, Rev1-D 0.152 s.

What this shows for the Discussion:

- **Rev1-D gives the most torque for its copper loss.** Km is 2.18 × Rev0, against 1.51 × for Rev1-S. K is 3.09 × Rev0 for 1.77 × the mass, and Rev1-D reaches 1000 rpm fastest (0.142 s).
- **Torque per magnet is about the same in all designs:** 1.83 / 1.95 / 1.88 mN·m per A_pk per magnet. Rev1 gains by adding magnet faces and a second stator, not by using each magnet better. Rev1-S is slightly best per magnet, because R1′s bare steel boosts S2.
- **The cost is speed.** At the same 12 V bus, Rev1-D reaches exactly 1000 rpm at 20 mN·m, so it has no speed headroom at the load point. Restoring headroom needs a higher bus voltage or fewer series turns. Connecting S1 and S2 in parallel is not an option for Rev1-D, because their EMFs differ by a factor of about two and current would circulate between them.
- **Efficiency figures count copper loss only, so they are upper bounds.** Iron loss in the steel and eddy loss in the copper would lower them, mostly in Rev1-D, which has the most steel and the highest field.
- **Inductance is insensitive for average performance** at these speeds (ωe·L ≈ 1 % of R).

Using this record in the report

The Methodology chapter can follow the validation chain in the order it was built: gates, the single board, the board-by-board stack, the full-motor model and its gate, then MATLAB. The Results chapter quotes the final values in "Final electromagnetic results" and "MATLAB system model". The Discussion draws on the agreement between methods, the boundary finding and the Rev1 trade-off.

Report section	What to write	Source in this record	Suggested figure or table
Methodology: tools and workflow	AEDT 2025 R2 driven by a scripted, checkpointed IronPython pipeline; build → verify → freeze → test on copies; Olwethu's decisions vs Claude's execution	How to use this record; Project setup; Engineering decisions	Pipeline stage diagram; version table
Methodology: geometry	Native ODB++ build, terminal choice, five-board stack with bridges, region rules	Timeline Phases 0, 1, 5; failures 1, 12–14	Stack cross-section with the axial levels
Methodology: electrical gates	Single-board conduction, inductance, FR-4 modelled as air	Timeline Phase 3; Rev0 validated results	Table of R per phase vs reference
Methodology: torque and back-EMF	Magnetostatic rotor-angle sweeps, torque reciprocity, copper torque, quality checks	Rev0 validated results (method, single board)	k(θ) waveform; quality-check table
Methodology: reduced-order stack	Coreless superposition, distance law, phasor stack sum, junction test	Rev0 validated results (distance law, boundary note, junction)	K1 vs distance

Report section	What to write	Source in this record	Suggested figure or table
Methodology: three-phase validation	Single-phase superposition replacing the crashed three-phase model	Rev0 validated results (three-phase check)	Phase table
Methodology: full-motor model	Field-only FEM with trace-resolved winding post-processing: why the copper can be omitted, trace extraction, the Lorentz sum, the 2 % gate	Full-motor field-only FEM and the validation gate; Timeline Phases 10–11	Flow diagram (Maxwell field → trace points → torque → k(θ)); plot of one board's trace paths
Methodology: verification of mesh and region	Mesh convergence in both models; the B·n = 0 region boundary and its analytic model; 20 mm padding	Full-motor section (gate history, convergence table); Timeline Phases 12–13; failure 34	Gate-history table; K1 vs padding chart
Methodology: Rev1	Two designs; two-rotor superposition with FEM factor f; steel-image coefficient c; S2 polarity from the FEM phase	Rev1: two designs	Rev1 topology sketch
Methodology: system model	Lumped model, parameter provenance, three models, how v2 takes K from the full-motor model	MATLAB system model	Equations; parameter table
Results: Rev0	K, T, E, Km, R, L; gap and back-plate DOE; per-board shares	Final electromagnetic results; Rev0 validated results	DOE table; per-board bar chart

Report section	What to write	Source in this record	Suggested figure or table
Results: Rev1	K per stator and machine; S2 polarity; plate field; rotor attraction force	Final electromagnetic results; per-board table; Rev1 board-by-board table	Per-board K for all five stators
Results: dynamics and comparison	Final MATLAB comparison; start-up, ripple, envelopes, efficiency	MATLAB system model	Comparison table; torque–speed envelopes; efficiency maps
Discussion	Two methods agree on ratios within 2 %; the region-boundary finding; torque per magnet; Km gain vs speed loss; Rev1-D has no speed headroom at 12 V; copper-loss bound; Rev1 S2 caveat	Final electromagnetic results; Full-motor section; MATLAB (what this shows)	Relative figures-of-merit chart
Conclusions and recommendations	The 16 GB limit applies to meshed copper only; the field-only model removes it; always check region size; prototype tests; thermal model	Full-motor solves and the computational limit; Open items	RAM vs model-size table

Points worth making explicitly in the report:

- **Every method was validated, not assumed.** Copper-torque linearity is 0.005 %, the three-phase superposition is within 0.11 %, and the distance law fits within 0.17 %. The full-motor model meets its gate at +1.40 % and 0.04°.
- **Two independent methods agree on the design comparison.** The board-by-board method and the full-motor model agree on Rev1/Rev0 within 2 %, and both find S2 in phase with S1.

- **The gate caught a real modelling bias.** The region boundary 5 mm from a coreless stator suppressed K by about 3–4 % at 1 mm, and more at larger distances. Mesh refinement could not reveal it, because both meshes had converged. The size of the solution region must be checked like the mesh.
- **The failed runs are evidence.** They mark exactly where 16 GB stops for meshed copper: three phases plus the rotor on one board, the inductance-matrix step, and the full five-board motor. The field-only model runs a full 48-magnet motor in about 12 min.
- **Every number carries its provenance** ([AEDT] / [Py] / [Est] / [Assump]). L, J and the drive values are estimates or assumptions; K, R and the field come from FEM.
- **Wording rules from the review:**
  - Say "field-only FEM with trace-resolved winding post-processing", never "exact".
  - Say "copper-loss-limited efficiency".
  - Label voltages as phase peak, rms, line-to-line or bus.
  - State the speed controller.
  - Say inductance is "insensitive for average performance", not irrelevant.

Open items

The electromagnetic modelling is complete as of 05:52 on 8 Oct. Still open: one Rev1 consistency check, one build, two confirmations from Olwethu, archiving, the prototype tests and the separate thermal model.

- [x] ~~REV1D_S2_K3 finished (about 01:40): f = 1.0726~~
- [x] ~~REV1D_S2_K1 finished (about 04:00) after the 23:44 freeze and the 02:02 crash: f = 1.0781. The cause of both failures was not determined~~
- [x] ~~REV1_S2_ENV finished (about 01:40): c = 1.166~~
- [x] ~~stack_sum.py: Rev0 / Rev1-S / Rev1-D table, S2 polarity, R1 plate field, axial force on R2~~
- [x] ~~S2 polarity set from the FEM phase: same sense as S1 (S2_REVERSED = False)~~
- [x] ~~Full-motor field-only model built, run for all three designs, and its gate passed (05:50)~~
- [x] ~~MATLAB final run with v2 (05:36)~~
- [ ] Rev1 S2 consistency check: tried on 8 Oct with a 1.0 mm gap mesh, and both runs froze; stopped at 08:37. If it is ever resumed, use run_REV1D_S2_K3_P20.py (20 mm padding, original mesh, about 2 h) and compare it with field-map S2 board 3 (7.5412 mN·m/A)
- [ ] Build the Rev1-S final geometry (run_REV1_FINAL.py, build only, about 40 min)
- [ ] Olwethu: confirm the "through" magnetisation for Rev1-D. All Rev1-D results use it
- [ ] Olwethu: confirm the magnet plan (Rev0 tested as S1 + R2; 48 magnets cover all builds)
- [ ] Choose the star-point ends (polarity is proven by the three-phase check)

- [ ] Archive in Motor run\results: MATLAB_REV0_FINAL, the v2 results_compare folder, FIELDMAP_*_P20 and REV0_DOE_D100_P20
- [ ] Pipeline: save step progress to OneDrive so that a lab-PC wipe cannot lose it
- [ ] Update the earlier Docs report ("Rev0 PCB Axial-Flux Motor: Maxwell Modelling, Optimisation and Final Build Report") with the final values
- [ ] Prototype tests. Back-EMF at a known speed against 1.444 V rms per 1000 rpm per phase (Rev0); phase resistance against 1.389 Ω plus leads; torque constant. For Rev1, design for the 47–48 N rotor attraction
- [ ] Thermal model (separate workstream since about 04:30 on 8 Oct). It takes the copper loss at the operating points from the MATLAB model
- [ ] Optional: an armature-reaction check (field with the winding current) at the 2 A_pk limit, and more rotor angles to resolve the 5th harmonic

Limitations to state in the report:

- The field-only model samples B on the trace centre-lines. It reads +1.4 % against the detailed reference, and it neglects armature reaction and eddy, skin and proximity effects in the copper.
- The Rev1 S2 boards were not gated directly. The field map reads 3–11 % above the detailed Rev1 single-board runs, and the cause is open.
- The trace-network R_dc (0.2656 Ω) is 4.4 % below the meshed-copper FEM value. The FEM value, 0.2777 Ω, is used everywhere.
- The rotor attraction force is a Maxwell-stress estimate on one mid-gap surface.
- The five-board inductance is an estimate (18.7–31.1 µH).
- The stack-up is the scaled Altium one, not the actual JLCPCB stack-up.
- Vias are modelled as solid 0.30 mm copper, so R comes out about 2.9 % high.
- The shaft material is assumed.
- 5th and higher harmonics are not resolved by 8 samples per period.
- The ±0.5 A linearity check and the magnet-radius (27/28/29 mm) study were not run.
- Efficiencies count copper loss only.

Times not recorded:

- the end of the Gate 1A solve;
- the start of the first three-phase try, the junction test, D050, PHB/PHC, and full attempts 1 and 2;
- the end of the v7 build;
- when +Y padding was fixed by hand;
- when the reviewer's endorsement of option 1 arrived (between 01:12 and 01:36 on 8 Oct);

- the exact finish of K3 and S2_ENV (both reported at 01:40);
- the start of FIELDMAP_REV0 and FIELDMAP_REV1D (after 02:10), REV0_DOE_D100_P20 and FIELDMAP_REV0_P20 (about 04:35).

The exact code change for the Optimetrics Analyze failure is not documented.